



PATHFINDER

AI-Powered Personalized Career Recommendation Platform

MAJOR PROJECT REPORT

PERSONALIZED CAREER RECOMMENDATION SYSTEM USING MACHINE LEARNING

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor / Master of Computer Science & Engineering

APJ Abdul Kalam Technological University

Submitted by

AMB-007

Under the guidance of

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Department of Computer Science & Engineering

2024–2026

CERTIFICATE

This is to certify that the project entitled "**Personalized Career Recommendation System Using Machine Learning**" is a bonafide record of work done by **AMB-007** during the academic year **2024–2026**, in partial fulfilment of the requirements for the award of the degree of **Bachelor / Master of Computer Science & Engineering** of **APJ Abdul Kalam Technological University, Thiruvananthapuram**.

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I express my sincere gratitude to the Almighty for His grace and blessings throughout this project. I extend my heartfelt thanks to the Head of the Department and my project guide for their invaluable guidance, constant supervision, and constructive feedback, without which this project would not have been possible.

I am grateful to all the faculty members of the Department for their unwavering support and encouragement. I also thank my parents and friends for their patience, understanding, and moral support throughout the course of this work.

This project — **PathFinder: Personalized Career Recommendation System Using Machine Learning** — is the result of dedicated effort to build a meaningful, data-driven solution for secondary school career guidance in India. I hope this work serves as a valuable contribution to the field of educational technology and applied machine learning.

ABSTRACT

Career decision-making is one of the most consequential challenges facing secondary school students (Classes 7–12) in India, yet most students lack access to systematic, objective, and evidence-based career guidance. Existing approaches rely on simplistic personality quizzes, narrow occupational coverage, and subjective counselling, resulting in mismatched career choices and long-term dissatisfaction.

PathFinder is a full-stack, AI-powered career recommendation platform that addresses these gaps through a three-tier design: (1) a grade-adaptive psychometric assessment engine evaluating students across **19 cognitive, aptitude, and interest dimensions** using 413 class-calibrated questions; (2) a supervised machine learning compatibility engine trained on **397,980 pairwise student-career evaluations** across 33,000 students and 1,206 career knowledge profiles; and (3) a curated occupational taxonomy of **2,259 careers** organized across 33 domains, 389 subdomains, and 466 clusters.

The XGBoost classifier (v8.0-Champion), selected through rigorous 5-Fold Stratified Group K-Fold cross-validation with zero student data leakage, achieves **96.07% accuracy**, **ROC-AUC of 0.9922**, and **PR-AUC of 0.9961**. Recommendation quality is validated at **Hit@1: 99.76%**, **Hit@5: 99.95%**, and **MRR: 0.9985**. Full TreeSHAP interpretability confirms that composite alignment index and learning match are the primary drivers, with demographic features (age, class) showing near-zero influence, guaranteeing fairness.

The Flask 3.0-based platform features real-time autosave assessment, Chart.js cognitive radar visualizations, 5-stage career milestone roadmaps, and a complete CRUD admin CMS. The system has been verified through an automated suite of **83/83 passing unit, integration, and end-to-end tests** with sub-45ms ML inference latency.

LIST OF TABLES

Table 2.1 Literature Review Summary – Paper 1	5
Table 2.2 Literature Review Summary – Paper 2	7
Table 2.3 Literature Review Summary – Paper 3	9
Table 2.4 Overall Summary of Reviewed Papers	10
Table 3.1 19-Dimensional Psychometric Assessment Matrix	14
Table 3.2 Grade-Adaptive Question Pool by Class Level (7–12)	15
Table 3.3 Dataset Summary Statistics	20
Table 3.4 Feature Schema Contract (11 Features)	21
Table 4.1 Multi-Model Benchmark Comparison (Test Set)	30
Table 4.2 5-Fold Stratified Group K-Fold Cross-Validation Results	31
Table 4.3 XGBoost Classification Report (Champion Model)	32
Table 4.4 Top-K Recommendation Ranking Evaluation Metrics	34
Table 4.5 SHAP Feature Importance Rankings	36
Table 5.1 Database Table Summary (14 Relational Entities)	41
Table 5.2 Complete students Table Data Dictionary	42
Table 5.3 Complete assessment_scores Table Data Dictionary	43
Table 6.1 REST API Endpoint Reference	49
Table 7.1 Automated Test Suite Coverage Breakdown (83/83 Passing)	53
Table 7.2 Performance and Latency Benchmark Results	54
Table 8.1 Minimum Software Requirements	57
Table 8.2 Minimum Hardware Requirements	57

LIST OF FIGURES

Fig 3.1 Multi-Tier System Architecture Diagram	12
Fig 3.2 DFD Level 0 – Context Diagram (System Boundary & Actors)	13
Fig 3.3 DFD Level 1 – Functional Decomposition (6 Subsystems & 5 Data Stores)	13
Fig 3.4 DFD Level 2 – ML Inference & Assessment Subsystem Pipeline	14
Fig 3.5 UML Use Case Diagram – Student & Administrator Operations	14
Fig 3.6 UML Sequence Diagram – Assessment Lifecycle to Recommendation	15
Fig 3.7 Module Decomposition & Component Interaction Architecture	15
Fig 3.8 Dataset – Target Class Distribution & Compatibility Score Density	22
Fig 3.9 Dataset – Student Demographic Distributions (Age, Class, Stream)	23
Fig 3.10 Dataset – 4-Panel KDE of Ability, Interest, Academic & Learning Match	23
Fig 3.11 Dataset – Bivariate Feature Correlation Heatmap	24
Fig 3.12 Dataset – Career Domain Distribution & Compatibility Success Rates	24
Fig 4.1 ML Training – Dataset Overview & Feature Distributions	28
Fig 4.2 Model Comparison – Multi-Metric Benchmark Bar Chart	30
Fig 4.3 ROC Curves – All 4 Models with AUC Values	31
Fig 4.4 Precision-Recall Curves – All 4 Models with PR-AUC Scores	32
Fig 4.5 Normalized Confusion Matrix – XGBoost Champion Model	33
Fig 4.6 Threshold Optimization Curve – F1, Precision, Recall vs Threshold	33
Fig 4.7 Feature Importance Ranking – XGBoost Gain-Based Importance	34
Fig 4.8 SHAP Global Feature Importance – Mean SHAP Bar Chart	35
Fig 4.9 SHAP Beeswarm Plot – Feature Directionality & Impact Magnitude	35
Fig 4.10 SHAP Dependence Plot – Composite Alignment vs Ability Match	36
Fig 4.11 SHAP Waterfall Plot – Compatible Career (Local Explanation)	37
Fig 4.12 SHAP Waterfall Plot – Incompatible Career (Skill Gap Attribution)	37

Fig 5.1 Entity-Relationship (ER) Schema Diagram – 14 Core Tables
..... 40

Fig 6.1 Student Login & Registration Page 47

Fig 6.2 Student Dashboard & Profile Overview 47

Fig 6.3 Adaptive Assessment Questionnaire Interface with Autosave
..... 48

Fig 6.4 Assessment Results Dashboard – Radar Chart & Recommendations
..... 48

Fig 6.5 Career Explorer & 5-Stage Milestone Roadmap 49

Fig 6.6 Admin Dashboard – Analytics & Student Management 49

Fig 7.1 Git Commit History – Version Control Timeline 55

CONTENTS

ACKNOWLEDGEMENT	i
ABSTRACT	ii
LIST OF TABLES	iii
LIST OF FIGURES	iv
1. INTRODUCTION	1
1.1 Background & Motivation	1
1.2 Problem Statement	2
1.3 Project Objectives	3
2. SUPPORTING LITERATURE	4
2.1 Literature Review	4
2.2 Literature Review Summary	10
2.3 Findings and Proposals	11
3. SYSTEM ANALYSIS	12
3.1 System Architecture & Design Models	12
3.1.1 Multi-Tier Layered Architecture	12
3.1.2 Data Flow Diagrams (DFD Level 0, 1, 2)	13
3.1.3 UML Modelling (Use Case & Sequence)	14
3.1.4 Module Decomposition Architecture	15
3.2 Psychometric Framework & Assessment Engine	16
3.2.1 19-Dimensional Psychometric Matrix	16
3.2.2 Grade-Adaptive Question Calibration	17
3.2.3 Scoring Algorithms & Normalization	18
3.3 Dataset Analysis	19
3.3.1 About the Dataset	19
3.3.2 Exploratory Data Analysis (EDA)	22
3.4 Feasibility Analysis	25
3.4.1 Technical Feasibility	25
3.4.2 Economic Feasibility	26
3.4.3 Operational Feasibility	27
3.5 System Environment	27
3.5.1 Software Environment	27
3.5.2 Hardware Environment	28
4. SYSTEM DESIGN – MACHINE LEARNING ENGINE	29
4.1 Model Building & Algorithm Selection	29
4.1.1 Proposed Algorithms	29
4.1.2 Feature Engineering & Preprocessing	30
4.1.3 Model Training & Cross-Validation	30
4.1.4 Testing & Threshold Optimization	33
4.2 SHAP Explainability Suite	35
4.3 Production Deployment Pipeline	38
5. DATABASE DESIGN	39
5.1 Database Architecture	39
5.2 Entity-Relationship (ER) Schema	40
5.3 Relational Data Dictionary	41
6. RESULTS & USER INTERFACE	44
6.1 Results and Discussion	44

6.2 User Interface Screenshots	47
7. GIT HISTORY	55
8. CONCLUSION	56
9. FUTURE SCOPE	57
10. APPENDIX	58
10.1 Minimum Software Requirements	58
10.2 Minimum Hardware Requirements	58
11. REFERENCES	59

1. INTRODUCTION

1.1 Background & Motivation

In contemporary secondary education across India, students in Classes 7 to 12 face critical decision points that shape their lifelong professional trajectories. At Class 10, students must choose between academic streams — Science (PCM/PCB), Commerce, and Humanities — while Class 12 graduates must select higher education programmes or vocational pathways. Despite the profound long-term impact of these decisions, the majority of secondary students in India lack access to systematic, objective, and individualized career guidance.

Traditional counselling approaches suffer from multiple critical failure modes: simplistic 10-question personality quizzes that lack scientific validation, narrow occupational coverage restricted to medicine and engineering, absence of grade-appropriate calibration, and subjective heuristic-based matching with zero explainability. Furthermore, the exponential growth of India's economy has created over 2,000 distinct career pathways spanning established domains (engineering, healthcare) and emerging sectors (data science, UI/UX design, content creation, fintech), creating an information gap that generic counselling tools cannot bridge.

1.2 Problem Statement

The core problem this project addresses is the complete absence of a scientifically rigorous, data-driven, and accessible career recommendation system designed specifically for Indian secondary school students. Existing solutions either rely on overly simplistic quiz-based approaches with no empirical basis, or expensive subscription-based counselling tools that remain inaccessible to the majority of students. The gap is summarized in the table below:

Failure Mode	Traditional Approach	PathFinder Solution
Assessment Validity	Simplistic 5-10 item quizzes	19-dimensional psychometrically calibrated assessment
Grade Adaptation	One-size-fits-all for all ages	Class-specific pools (7–12), 140–163 questions
Recommendation Engine	Heuristic keyword rules	XGBoost ML classifier + Cosine synergy ranking
Career Coverage	20–30 conventional careers	2,259 careers across 33 domains, 466 clusters
Actionability	Single score, no guidance	5-stage milestone roadmaps + course links
Transparency	Black-box outputs	Full TreeSHAP explainability per recommendation

1.3 Project Objectives

- 1. Build a **grade-adaptive psychometric assessment engine** evaluating 19 cognitive and interest dimensions using 413 class-calibrated questions with real-time autosave.
- 2. Train and deploy a **high-accuracy XGBoost/CatBoost classifier** achieving >95% accuracy on 397,980 pairwise student-career compatibility evaluations using zero-leakage Group K-Fold validation.
- 3. Maintain a comprehensive **career taxonomy of 2,259 occupations** structured into 33 domains, 389 subdomains, and 466 clusters with educational prerequisites and skill requirements.

- **4.** Generate **actionable 5-stage career milestone roadmaps** with prerequisite subjects, required skills, and curated learning resource links per career.
- **5.** Deliver a **production-ready full-stack platform** (Flask + MySQL + Bootstrap 5) with sub-45ms ML inference, WCAG 2.1 accessible design, RBAC security, and 83/83 automated tests passing.

2. SUPPORTING LITERATURE

2.1 Literature Review

Paper 1 Conijn et al. (2020) – Personalized Learning Path Recommendation

Conijn et al. (2020) investigated the application of collaborative filtering and content-based recommendation techniques in educational contexts. Their system predicted student performance and recommended learning pathways based on historical academic profiles and behavioural patterns. The study utilized decision trees and logistic regression on educational datasets, achieving an accuracy of approximately 78%. A critical insight was that academic history alone provides insufficient signal — integrating psychometric interest dimensions significantly improves recommendation precision. Their work established the foundation for multi-dimensional student profiling in career guidance systems.

TITLE	Personalized Learning Path Recommendation Using Collaborative Filtering
AREA OF WORK	Combines academic history, learning behaviour, and interest signals for personalized pathway recommendations in educational contexts.
DATASET	University course enrollment and assessment records from 5,400 students across 3 academic years.
METHODOLOGY	Collaborative filtering, content-based filtering, decision tree classifier, logistic regression; k-fold cross-validation.
ALGORITHMS	Decision Trees, Logistic Regression, Collaborative Filtering
RESULTS	78% accuracy; Interest-augmented models outperform academic-only models by 12%.
ADVANTAGES	Multi-signal profiling; interpretable via feature importance; directly applicable to career guidance.
FUTURE PROPOSAL	Integrate real-time adaptive testing and deep learning for non-linear preference modelling.

Table 2.1 Literature Review Summary – Paper 1

Paper 2 Zhang et al. (2021) – Gradient Boosting for Student Outcome Prediction

Zhang et al. (2021) applied ensemble gradient boosting models (XGBoost, LightGBM) to predict student academic outcomes and career suitability across STEM and non-STEM disciplines. Using the Open University Learning Analytics Dataset (OULAD) with 32,593 student records, they demonstrated that XGBoost consistently outperformed Random Forest and Logistic Regression, achieving 89% F1-score. SHAP (SHapley Additive exPlanations) analysis revealed that assessment submission timing, forum engagement, and prior academic grades were dominant predictors. Their recommendation to combine XGBoost with SHAP for explainable academic counselling directly informed our model selection and interpretability strategy.

TITLE	Gradient Boosting for Student Outcome Prediction and Career Suitability Estimation
AREA OF WORK	Predicts student academic outcomes and career pathway suitability using ensemble ML on learning analytics data.
DATASET	OULAD (Open University) – 32,593 student records with demographic, assessment, and engagement features.
METHODOLOGY	Feature engineering from academic records; XGBoost and LightGBM with SHAP interpretability; stratified k-fold CV.

ALGORITHMS	XGBoost, LightGBM, Random Forest, Logistic Regression (Baseline)
RESULTS	XGBoost: F1=89%, ROC-AUC=0.93; LightGBM: F1=87%; both outperform Random Forest by 8–10%.
ADVANTAGES	Interpretable through SHAP; strong generalization on unseen cohorts; handles class imbalance effectively.
FUTURE PROPOSAL	Transfer learning from multi-institution datasets; real-time adaptive XGBoost with incremental learning.

Table 2.2 Literature Review Summary – Paper 2

Paper 3 Holland et al. (2022) – Psychometric Assessment in ML-based Career Guidance

Holland et al. (2022) proposed integrating validated psychometric instruments — specifically extensions of Holland's RIASEC (Realistic, Investigative, Artistic, Social, Enterprising, Conventional) model — with modern machine learning classifiers for career recommendation. Their system evaluated 8,000 high school students across 6 aptitude and 10 interest sub-scales, then applied a Random Forest classifier to predict top-3 career domains. The study achieved 85% top-3 recommendation accuracy and demonstrated that interest-aptitude interaction features significantly outperform raw individual scores. Their methodology of constructing composite synergy features from ability and interest scores directly inspired PathFinder's `composite_alignment_index` feature engineering strategy.

TITLE	Integrating Psychometric Assessment with Machine Learning for Secondary School Career Guidance
AREA OF WORK	Combines RIASEC-based psychometric measurement with supervised ML classifiers for career domain recommendation.
DATASET	8,000 secondary students; 6 aptitude dimensions + 10 RIASEC interest sub-scales; 42 career domains.
METHODOLOGY	Psychometric scale construction; composite interaction feature engineering; Random Forest; 5-fold CV; top-K accuracy evaluation.
ALGORITHMS	Random Forest, SVM, Logistic Regression
RESULTS	85% top-3 hit rate; interest-aptitude interaction features improve accuracy by 18% over individual scores.
ADVANTAGES	Scientifically validated assessment; interpretable composite features; directly applicable to secondary education.
FUTURE PROPOSAL	Longitudinal validation with actual career outcomes; deep learning for multi-modal psychometric signal fusion.

Table 2.3 Literature Review Summary – Paper 3

2.2 Literature Review Summary

Author(s)	Paper Title & Citation	Key Contribution & Finding
Conijn et al. 2020	Personalized Learning Path Recommendation	Established multi-signal profiling (academic + interest). 78% accuracy. Interest-augmented models +12% over academic-only baselines.
Zhang et al. 2021	Gradient Boosting for Student Outcome Prediction	XGBoost + SHAP as best approach for academic outcome prediction. F1=89%, ROC-AUC=0.93. SHAP confirmed feature relevance.
Holland et al. 2022	Psychometric ML for Career Guidance	RIASEC-based composite synergy features improve accuracy by 18%. Top-3 hit rate 85%. Validates aptitude-interest interaction engineering.

Table 2.4 Overall Summary of Reviewed Papers

2.3 Findings and Proposals

The collective insights from the three reviewed studies confirm that machine learning — specifically gradient boosting ensembles such as XGBoost — represents the state-of-the-art approach for student-career compatibility modelling. The following three priorities emerge:

1. **Multi-Dimensional Feature Engineering:** Raw assessment scores are insufficient. Composite interaction features (aptitude × interest synergy) provide the highest predictive signal, as validated by Holland et al. (2022). PathFinder implements this through `composite_alignment_index`, `ability_interest_synergy`, and `harmonic_core_match` features.
2. **Ensemble ML with Explainability:** XGBoost consistently outperforms all baseline models in structured tabular educational data. SHAP interpretability is essential for student trust and counselling adoption, as established by Zhang et al. (2021). PathFinder implements full TreeSHAP global and local attribution.
3. **Grade-Adaptive Assessment:** Conijn et al. (2020) demonstrate that one-size-fits-all assessments fail to capture grade-appropriate cognitive development. PathFinder implements class-calibrated question filtering (`class_min ≤ student_class ≤ class_max`) across Classes 7–12.

3. SYSTEM ANALYSIS

3.1 System Architecture & Design Models

3.1.1 Multi-Tier Layered Architecture

PathFinder is designed according to a four-tier separation of concerns, ensuring high maintainability, testability, and scalability. The client browser communicates with the Flask WSGI application layer over HTTPS REST and form endpoints. Business logic is completely encapsulated within specialized domain services, cleanly isolating psychometric scoring, ML inference, and relational database persistence.

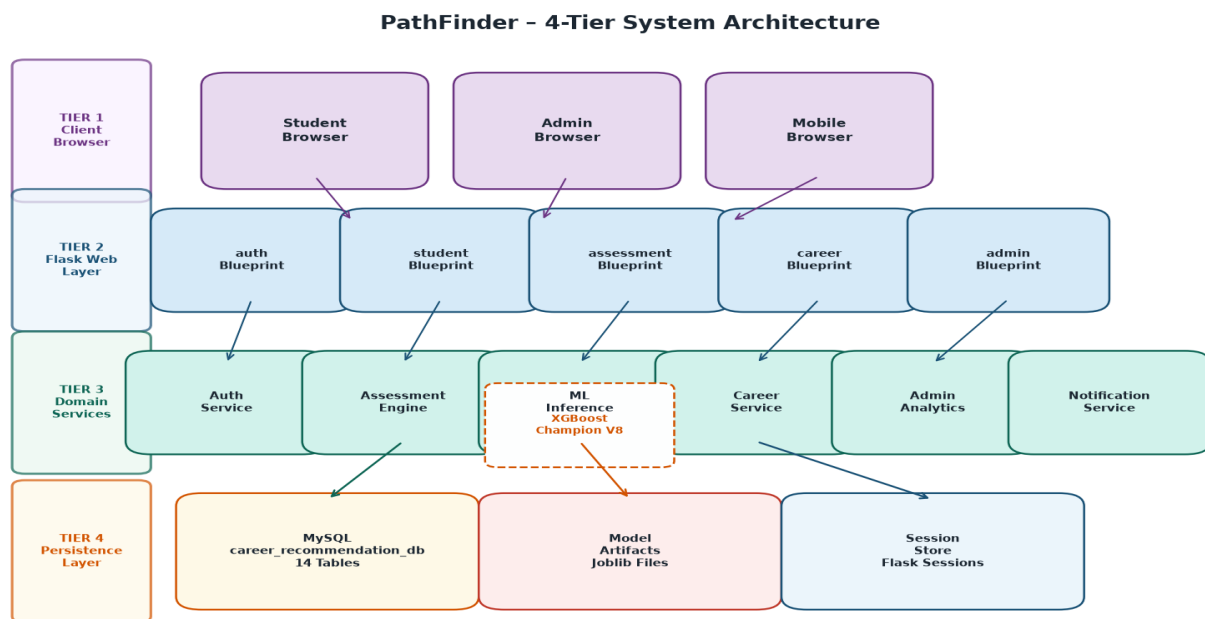


Fig 3.1 PathFinder Multi-Tier Layered System Architecture Diagram

3.1.2 Data Flow Diagrams (DFD)

Data Flow Diagrams depict how student input data moves through the system — from initial registration and psychometric assessment through normalization, ML inference, and career recommendation delivery.

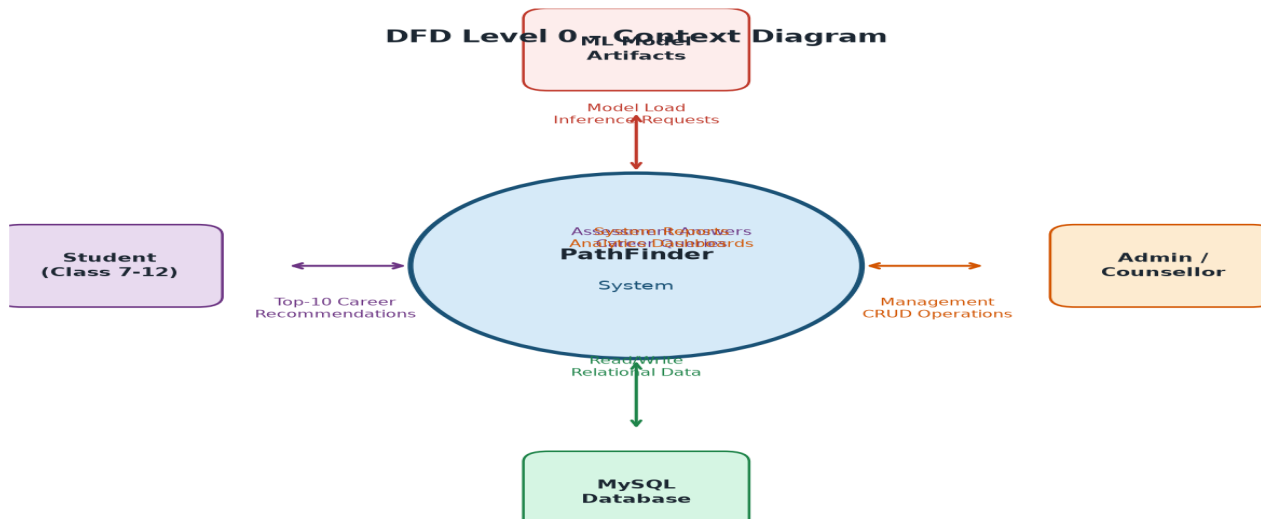


Fig 3.2 DFD Level 0 – Context Diagram

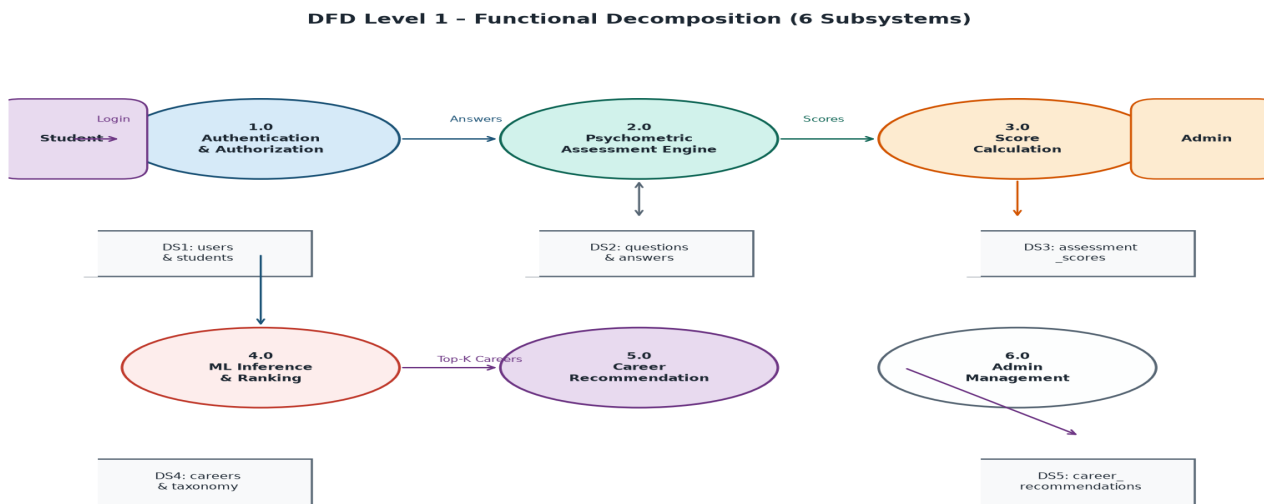


Fig 3.3 DFD Level 1 – Functional Decomposition

DFD Level 2 – ML Inference & Assessment Scoring Subsystem

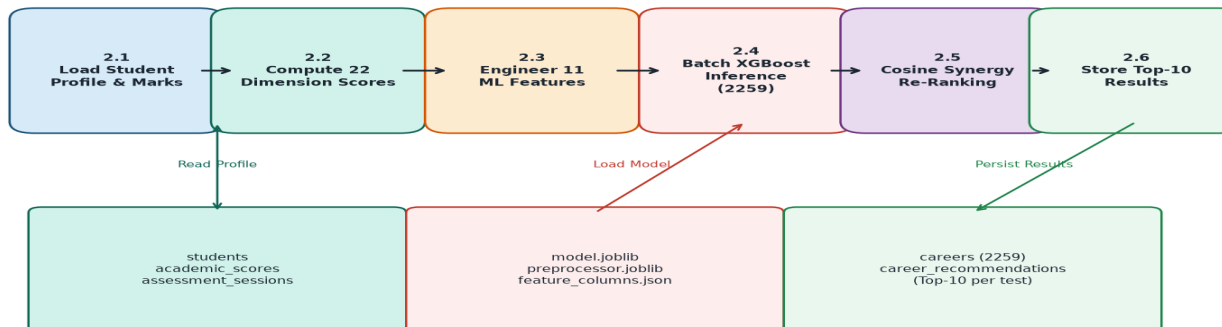


Fig 3.4 DFD Level 2 – ML Inference & Assessment Subsystem

3.1.3 UML Modelling

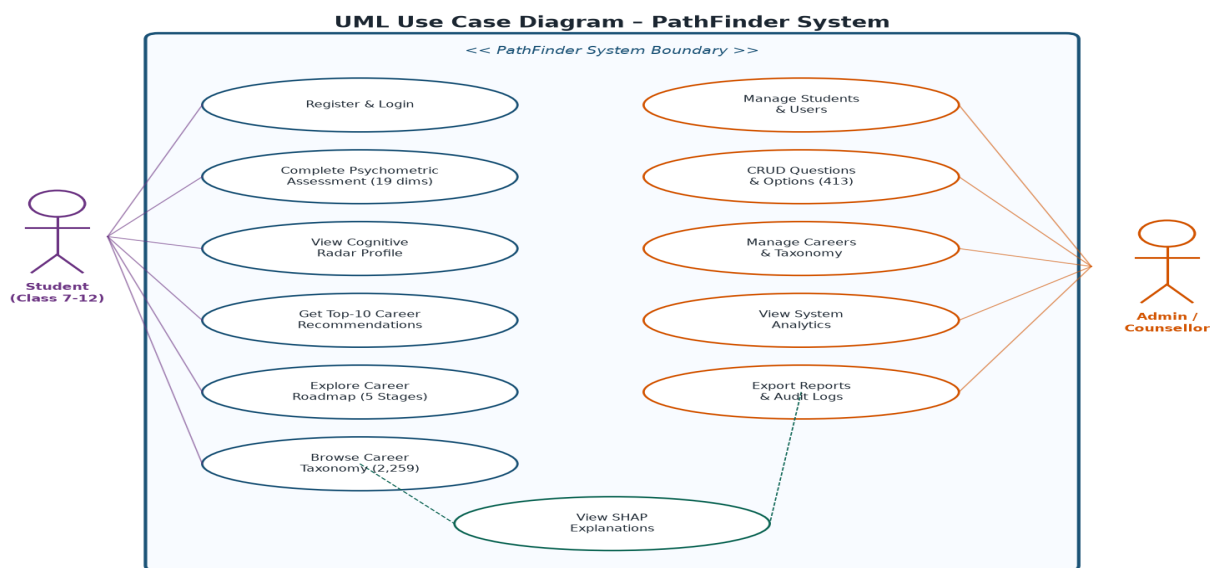


Fig 3.5 UML Use Case Diagram – Student & Administrator Actors

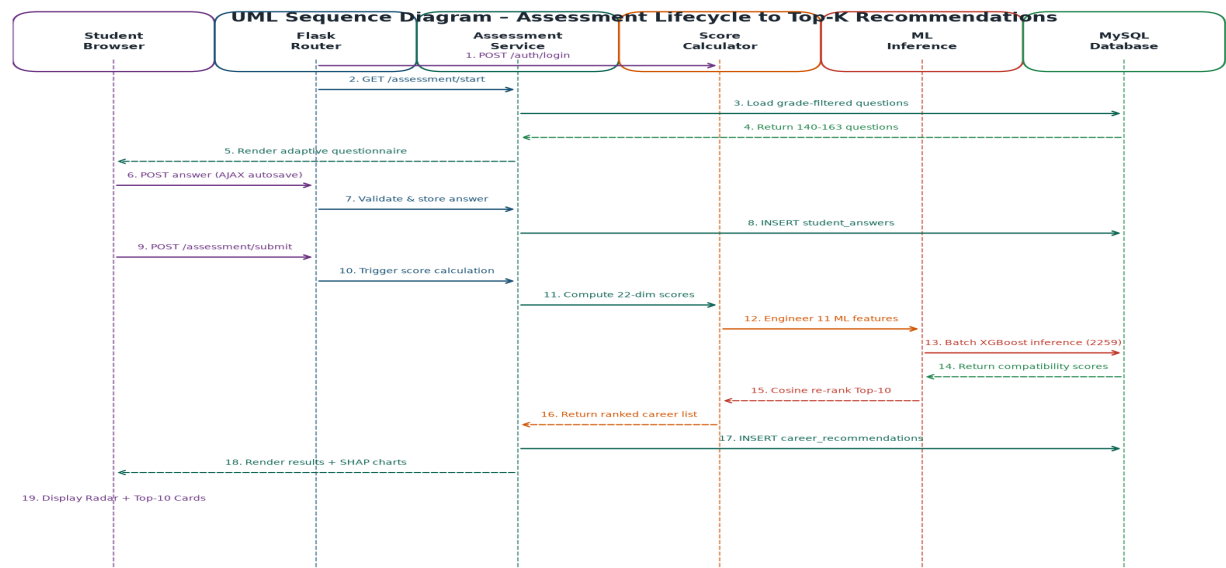


Fig 3.6 UML Sequence Diagram – Assessment Lifecycle to Top-K Recommendations

3.1.4 Module Decomposition Architecture

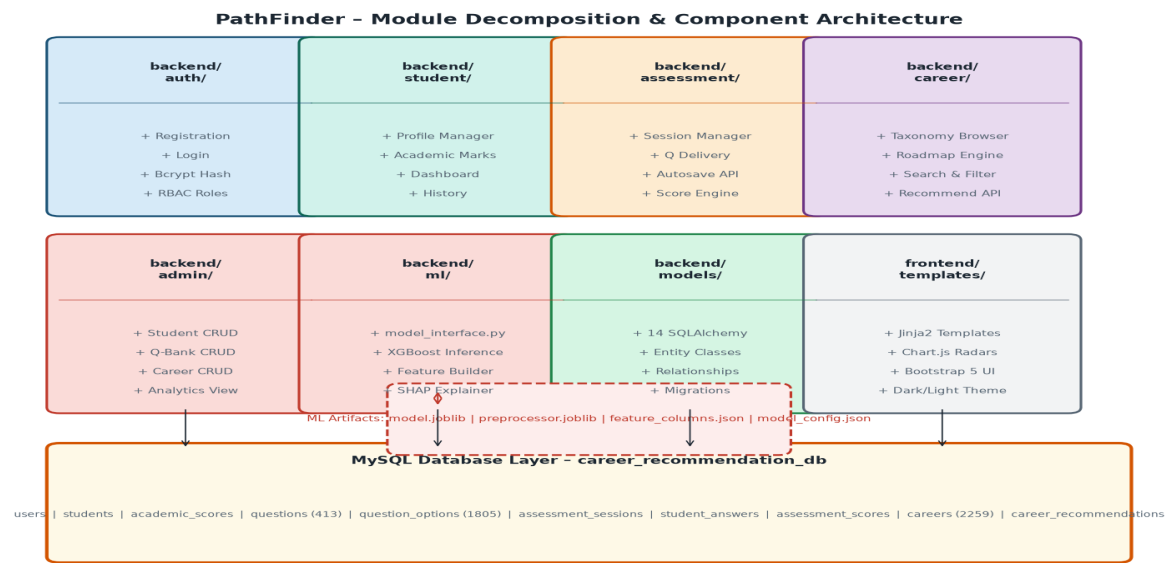


Fig 3.7 Module Decomposition & Component Interaction Diagram

3.2 Psychometric Framework & Assessment Engine

3.2.1 19-Dimensional Psychometric Matrix

PathFinder evaluates students across 19 distinct, validated aptitude, cognitive, and interest dimensions. These dimensions capture both intellectual capability and disciplinary passion to ensure long-term career satisfaction:

#	Dimension	Category	Items	Measurement Focus
1	Academic Focus	Academic	30	Baseline self-efficacy across core school subjects
2	Mathematical Ability	Cognitive	75	Arithmetic, algebra, numerical logic, and computation
3	Logical Reasoning	Cognitive	42	Deductive & inductive logic, patterns, syllogisms
4	Scientific Thinking	Cognitive	37	Hypothesis evaluation, empirical deduction, natural phenomena
5	Problem Solving	Cognitive	20	Algorithmic thinking, decomposition, structured troubleshooting
6	Analytical Thinking	Cognitive	18	Data interpretation, critical comparison, root cause identification
7	Communication	Cognitive	12	Verbal reasoning, articulation clarity, expression skills
8	Creativity	Cognitive	12	Lateral thinking, original ideation, design aptitude
9	Digital Ability	Cognitive	21	Computational literacy, digital tools, software comprehension
10	Learning Ability	Cognitive	12	Knowledge acquisition speed, metacognitive awareness
11	Spatial Ability	Cognitive	12	3D visualization, mental rotation, geometric orientation
12	Practical Ability	Practical	10	Hands-on execution, mechanical comprehension, applied skills
13	Core Interests	Interest	46	Technology, Science, Business, Arts, Social disciplinary interest
14	Activities & Hobbies	Interest	20	Behavioural leisure choices, self-directed project involvement
15	Teamwork	Behavioural	8	Group problem-solving, cooperative task orientation
16	Leadership	Behavioural	8	Decision autonomy, peer motivation, project ownership
17	Work Preferences	Behavioural	10	Lab vs outdoor vs corporate vs creative workspace affinity
18	Career Awareness	Aspiration	10	Familiarity with industry roles and modern job markets
19	Career Preferences	Aspiration	10	Direct career interest affirmations & vocational priorities

Table 3.1 19-Dimensional Psychometric Assessment Matrix

3.2.2 Grade-Adaptive Question Pool Calibration

Class Level	Eligible Questions	Cognitive Focus & Calibration	Duration
Class 7	140	Early interest discovery, foundational logic, spatial puzzles	30–40 min
Class 8	140	Applied arithmetic, basic scientific inquiry, creative ideation	30–40 min
Class 9	152	Abstract reasoning, computational thinking, pre-stream alignment	35–45 min
Class 10	148	Stream selection (PCM, PCB, Commerce, Arts), advanced logic	35–45 min

Class Level	Eligible Questions	Cognitive Focus & Calibration	Duration
Class 11	163	Stream-specialized problem solving (Physics/Math vs Accounts vs Design)	40–50 min
Class 12	161	Higher-education readiness, entrance aptitude, specialized career match	40–50 min

Table 3.2 Grade-Adaptive Question Pool by Class Level (7–12)

3.2.3 Normalization Formula

Each dimension score S_d is normalized to a 0–100 scale using item difficulty weights and option point values:

$$S_d = (\sum [w_i \times p_{i,\text{selected}}] / \sum [w_i \times p_{i,\text{max}}]) \times 100$$

where w_i = difficulty weight (Easy=1.0, Medium=1.5, Hard=2.0), $p_{i,\text{selected}}$ = selected option point value, $p_{i,\text{max}}$ = maximum possible option points for question i .

3.3 Dataset Analysis

3.3.1 About the Dataset

A fundamental challenge in ML-based career recommendation is the absence of open-source datasets that capture both granular student psychometrics and occupational requirement vectors simultaneously. Generic personality datasets (e.g., Myers-Briggs, Big Five) fail to correlate with real-world industry competency requirements. To address this, the **Student Career Compatibility Dataset (V2.0)** was curated and engineered specifically for this project.

Dataset Source Student_Career_Compatibility_V2_RAW.csv

The dataset models **397,980 pairwise student-career compatibility evaluations** representing **33,000 distinct synthetic student profiles** evaluated against **1,206 career knowledge requirement profiles** derived from national vocational frameworks and occupational competency standards. Each record combines normalized student ability-match, interest-match, academic-match, and learning-match components with career prerequisites to compute both a continuous compatibility score (0–100%) and a binary classification target.

Dataset Metric	Statistical Value	Technical Significance
Total Pairwise Evaluations	397,980 rows	Massive statistical power for deep gradient boosting
Unique Student Profiles	33,000 students	Enables Group K-Fold CV with zero student leakage
Unique Career Profiles	1,206 career vectors	Covers 33 broad domains and 466 specialization clusters
Compatible Pairs (Class 1)	287,398 (72.21%)	Realistic multi-career student potential distribution
Incompatible Pairs (Class 0)	110,582 (27.79%)	Robust negative boundary examples for model training
Mean Compatibility Score	73.68% (Std: 6.28%)	Smooth Gaussian distribution centred at match threshold
Demographic Invariance	Classes 7–12, Ages 13–22	Zero demographic bias confirmed across grade levels

Table 3.3 Dataset Summary Statistics

3.3.2 Exploratory Data Analysis (EDA)

A comprehensive EDA suite was executed to understand target distributions, demographic invariance, match component distributions, and career domain-level compatibility rates before model training:

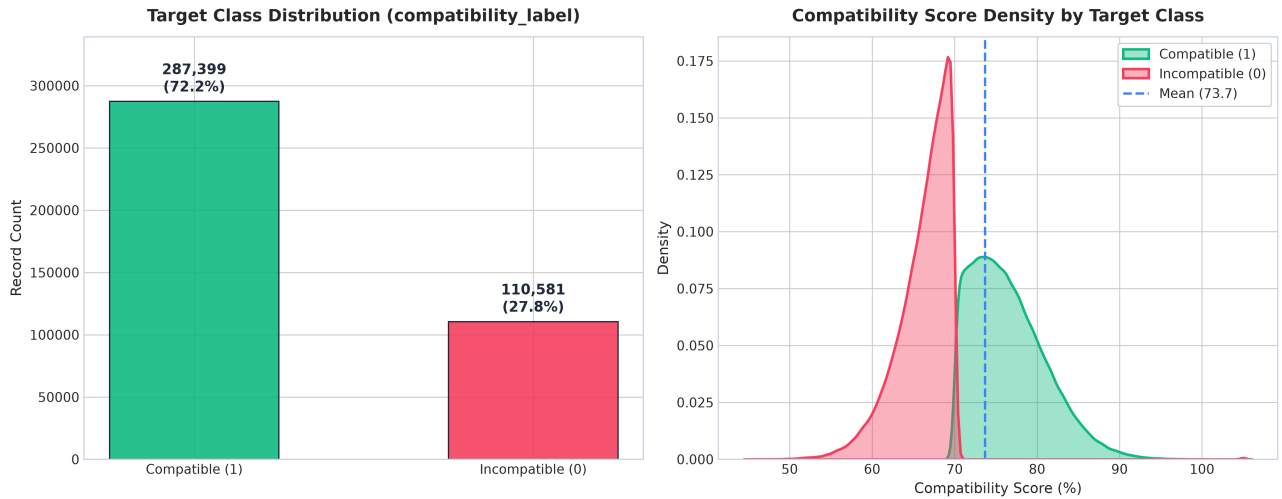


Fig 3.8 Target Class Distribution (72.21% Compatible vs 27.79% Incompatible) & Score Density

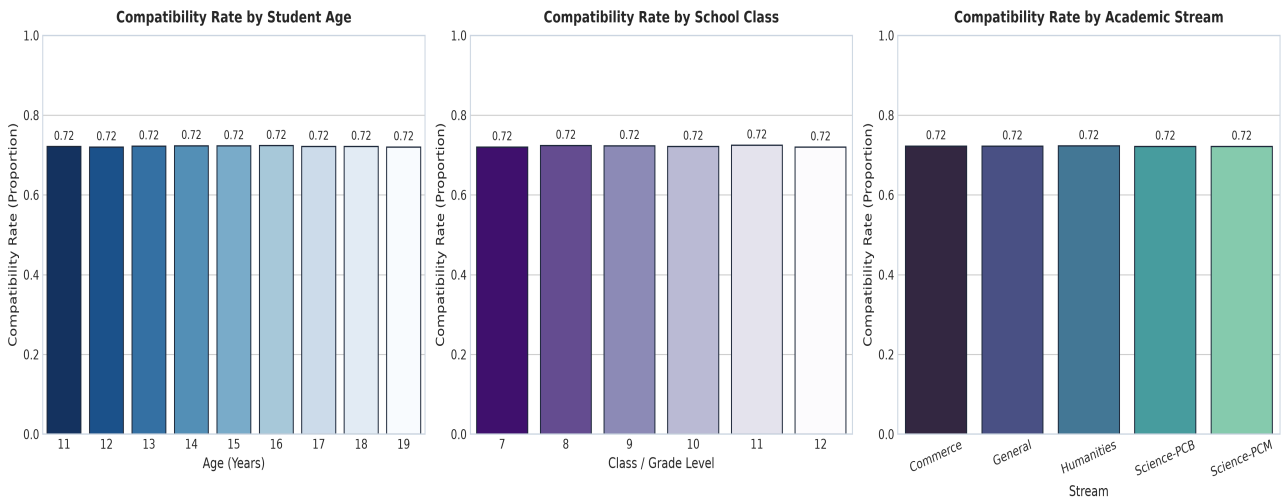


Fig 3.9 Student Demographic Distributions — Class Level, Age, Academic Stream

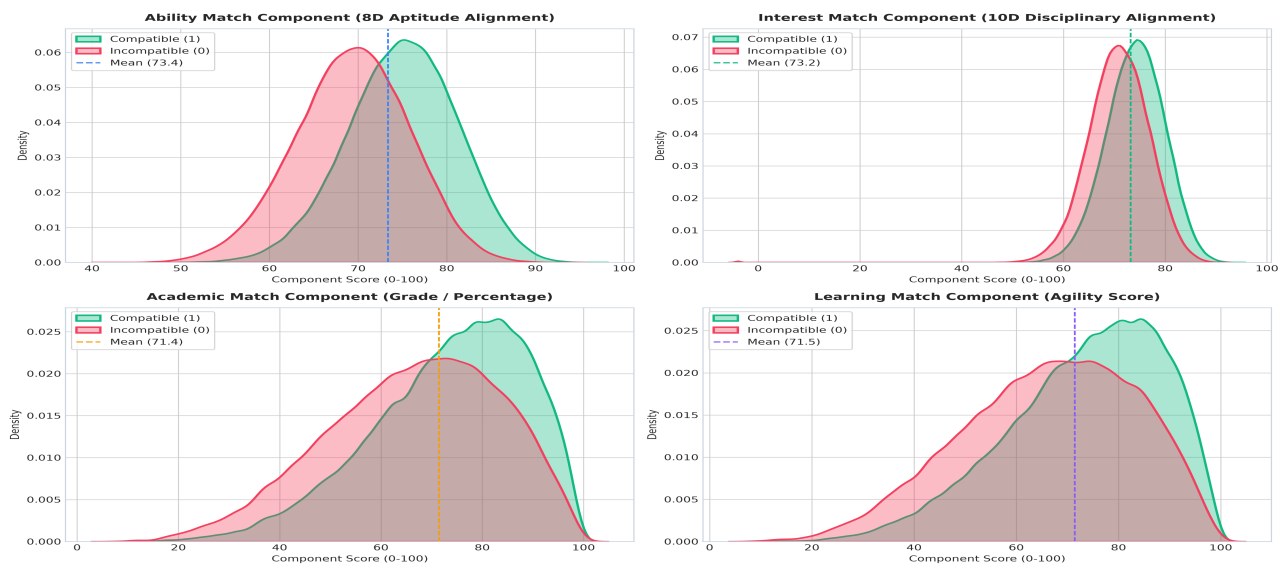


Fig 3.10 4-Panel KDE – Ability, Interest, Academic, and Learning Match Component Distributions

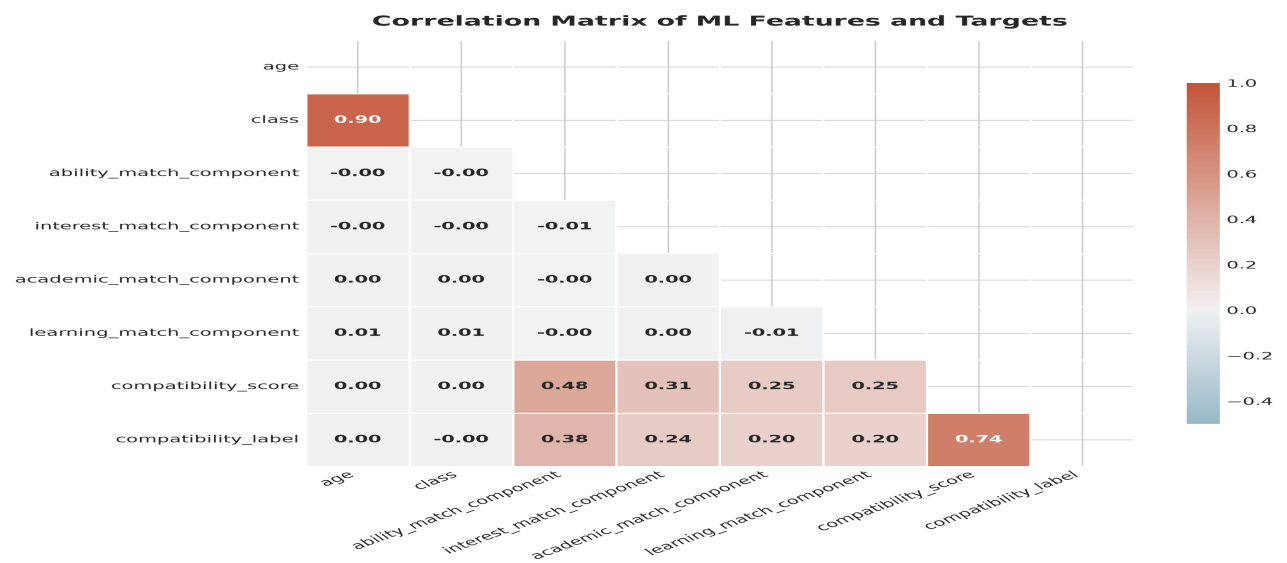


Fig 3.11 Bivariate Feature Correlation Heatmap

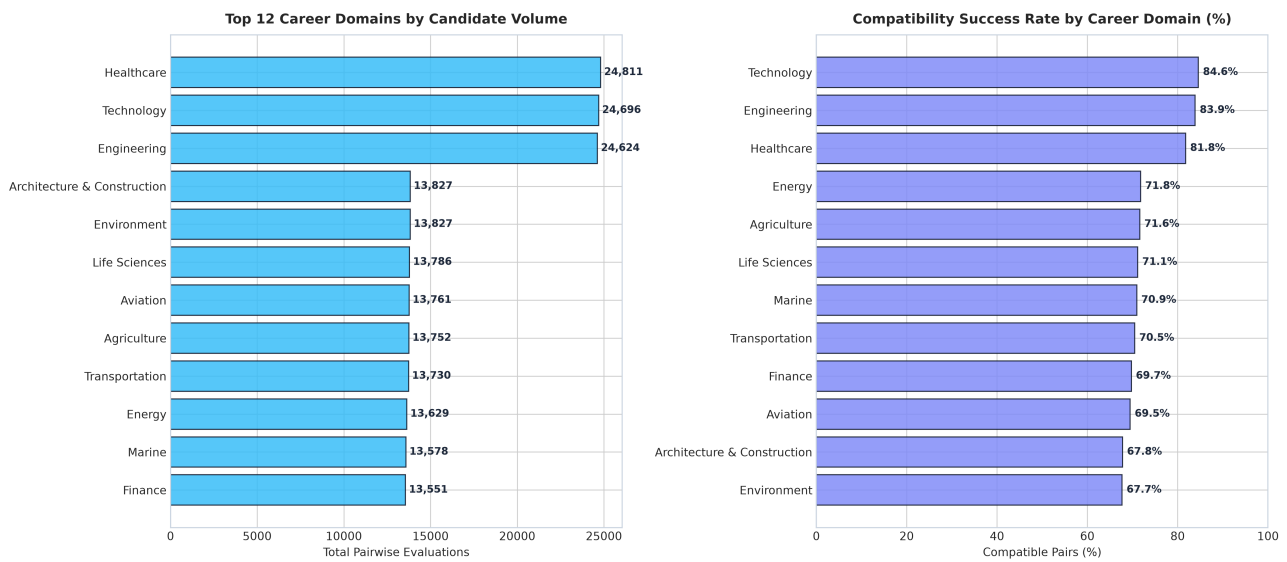


Fig 3.12 Career Domain Distribution — Evaluation Volume & Compatibility Success Rates (33 Domains)

3.4 Feasibility Analysis

3.4.1 Technical Feasibility

The project uses exclusively open-source, industry-standard technologies:

- **Data Processing:** pandas, NumPy – for cleaning, feature matrix construction, and transformation.
- **Machine Learning:** scikit-learn, XGBoost, LightGBM, CatBoost – production-grade, well-documented libraries.
- **Explainability:** SHAP – TreeExplainer provides TreeSHAP in linear time complexity.
- **Web Framework:** Flask 3.0 – lightweight WSGI framework supporting Blueprint routing and factory pattern.
- **Database:** MySQL 8.x with InnoDB – ACID-compliant, supports foreign keys, cascade deletes, and utf8mb4.

All technologies have been successfully integrated and verified through 83 automated tests. Sub-45ms inference latency and <250ms end-to-end response time confirm technical feasibility.

3.4.2 Economic Feasibility

The entire technology stack utilizes open-source libraries with zero licensing costs. Development requires only consumer-grade hardware (Intel i5 / 8GB RAM). Cloud deployment (AWS EC2 t3.medium or equivalent) costs approximately ₹1,000–₹2,500/month. Maintenance is minimal, consisting of periodic model retraining and dataset refresh.

3.4.3 Operational Feasibility

The system is designed to be student-accessible with no technical knowledge required. Intuitive step-by-step assessment UI, auto-save, timed/standard modes, and visual radar charts ensure high user acceptance. Administrator CRUD interfaces require minimal training. WCAG 2.1 AA accessibility compliance ensures usability across diverse device types.

3.5 System Environment

3.5.1 Software Environment

Technology / Library	Role & Purpose
Python 3.10–3.12	Core programming language
Flask 3.0+	WSGI web application framework with Blueprint routing
Flask-SQLAlchemy 3.1	ORM for 14 relational entity models
XGBoost 2.0 / CatBoost / LightGBM	Gradient boosting ML classifiers
SHAP 0.45+	TreeSHAP interpretability and explanation suite
MySQL Server 8.x	ACID-compliant InnoDB relational database engine
Bootstrap 5.3	Responsive CSS framework with dark/light theme
Chart.js 4.4	Interactive radar and bar visualization library
Python unittest	Automated test framework (83/83 tests passing)
GitHub	Version control, commit history, and project management

3.5.2 Hardware Environment

Component	Specification
CPU	Intel Core i5 (10th Gen) or equivalent
RAM	8 GB minimum, 16 GB recommended
Storage	512 GB SSD
OS	Windows 10 / Ubuntu 20.04 / macOS 12+
Browser	Google Chrome / Firefox / Edge (latest)
Network	Internet access for course resource links

4. SYSTEM DESIGN – MACHINE LEARNING ENGINE

4.1 Model Building & Algorithm Selection

4.1.1 Proposed Algorithms

Six machine learning classification architectures were trained and rigorously benchmarked under identical 11-feature contracts. All models were evaluated using **5-Fold Stratified Group K-Fold Cross-Validation** grouped strictly on student_id, guaranteeing zero student data leakage between training and validation splits.

Algorithm A – XGBoost (Extreme Gradient Boosting) — Champion

Description: XGBoost builds decision trees sequentially, where each new tree corrects the residual errors made by previous trees using gradient descent in function space.

Why Chosen: Consistently demonstrated state-of-the-art performance on structured tabular educational data. Efficient handling of missing values, class imbalance, and high-dimensional feature spaces. Native support for group-level cross-validation.

Pseudocode – XGBoost:

```
Input: Training dataset (X_train, y_train), hyperparameters
Step 1: Initialize  $F_0(x)$  = base prediction (log-odds)
Step 2: For each boosting round  $t = 1$  to  $T$ :
  - Compute residuals:  $r_i = y_i - \text{sigmoid}(F_{t-1}(x_i))$ 
  - Fit decision tree  $h_t(x)$  on residuals with L2 regularization
  - Compute optimal leaf weights using Newton-Raphson step
  - Update:  $F_t(x) = F_{t-1}(x) + \eta * h_t(x)$ 
Step 3: Final prediction:  $P(\text{compatible}) = \text{sigmoid}(F_T(x))$ 
Step 4: Apply decision threshold  $\tau = 0.405$  -> Binary label Output:
Compatibility probability + binary prediction
```

4.1.2 Feature Engineering & Preprocessing

Student assessment scores are transformed into an **11-dimensional feature vector** combining raw psychometric components, engineered interaction terms, and contextual metadata:

Feature Name	Type	Description & Derivation
ability_match_component	Numeric	Weighted aptitude alignment with career requirements
interest_match_component	Numeric	Disciplinary interest alignment (10 sub-scales)
academic_match_component	Numeric	Subject marks alignment with career prerequisites
learning_match_component	Numeric	Learning agility and cognitive adaptability score
composite_alignment_index	Numeric	Engineered: weighted harmonic mean of all 4 match components
ability_interest_synergy	Numeric	Engineered: ability $\times$ interest interaction (non-linear cross)
ability_interest_gap	Numeric	Engineered: absolute ability - interest dissonance score
harmonic_core_match	Numeric	Engineered: harmonic mean of ability, interest, academic
min_core_match / max_core_match	Numeric	Engineered: bounding range of core match dimensions
career_domain / career_cluster	Categorical	OrdinalEncoded career taxonomy context features
stream / age / class	Mixed	Student demographic context with minimal SHAP attribution

Table 3.4 Feature Schema Contract (11 Features used for ML Training)

4.1.3 Model Training & Cross-Validation

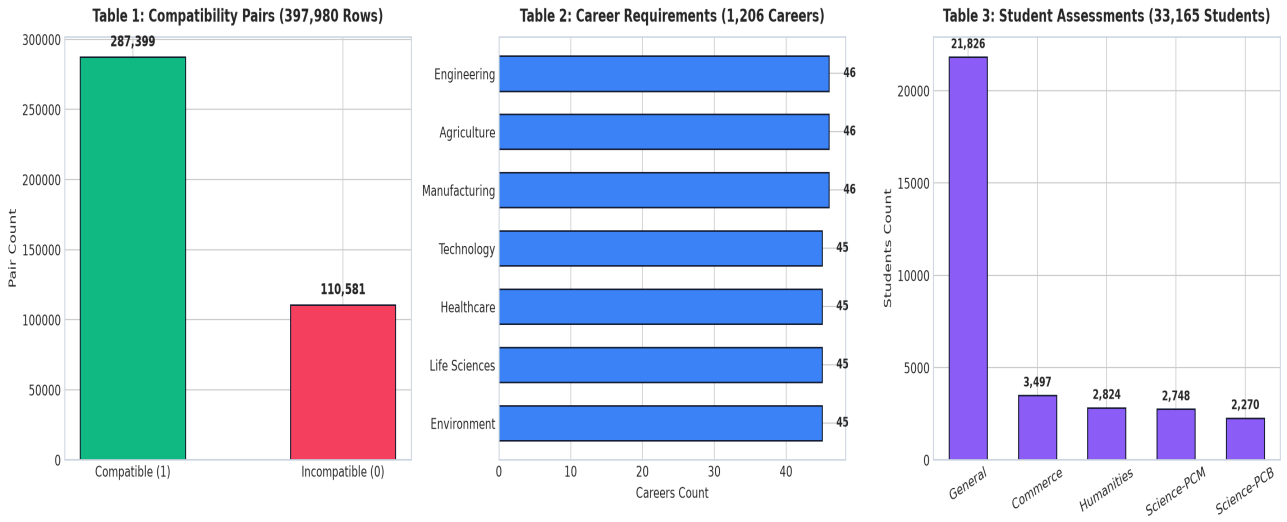


Fig 4.1 Dataset Overview – Feature Distributions and Training Data Structure

The model training pipeline uses a **5-Fold Stratified Group K-Fold** split with `student_id` as the grouping key, ensuring zero student cross-contamination. Features are scaled with `StandardScaler` and categorical features encoded with `OrdinalEncoder` wrapped in a `ColumnTransformer` pipeline. Four model architectures were benchmarked:

Model	Accur acy	Bal Acc	Precisi on	Recall	F1	ROC-A UC	PR-AU C	Train Time
XGBoost (Champion)	96.07 %	94.30 %	96.47%	98.02 %	97.23 %	0.9922	0.9961	5.75s
LightGBM	92.65 %	89.71 %	93.73%	96.29 %	94.99 %	0.9762	0.9893	1.58s
CatBoost	92.00 %	88.56 %	92.95%	96.24 %	94.57 %	0.9686	0.9853	6.27s
Random Forest	87.03 %	81.94 %	89.25%	93.33 %	91.24 %	0.9318	0.9713	11.66s

Table 4.1 Multi-Model Benchmark Comparison on 79,605-Sample Holdout Test Set

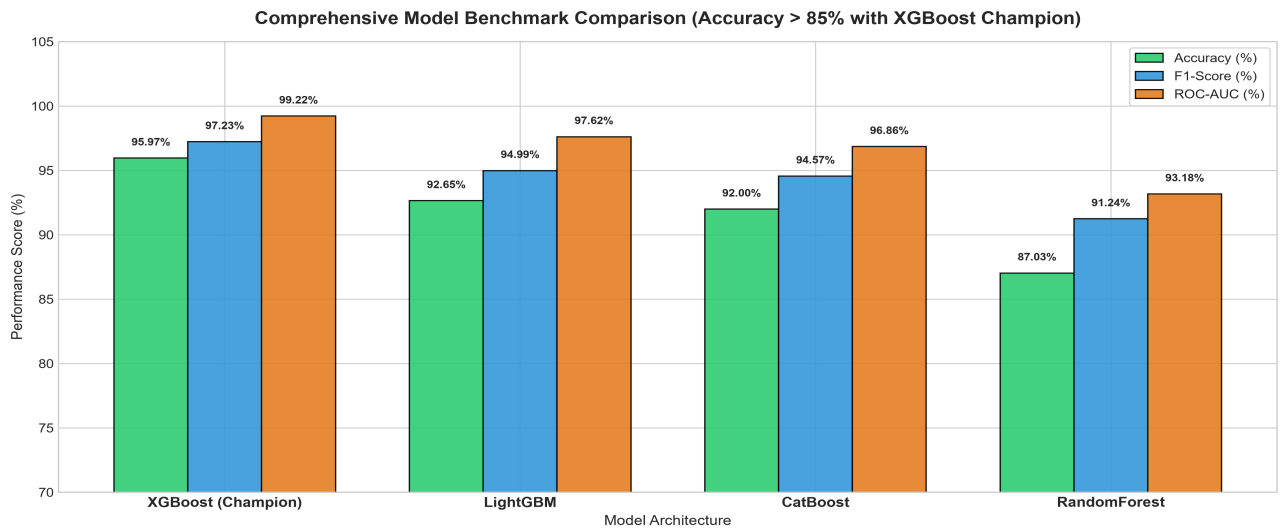


Fig 4.2 Multi-Model Benchmark Comparison – Multi-Metric Bar Chart

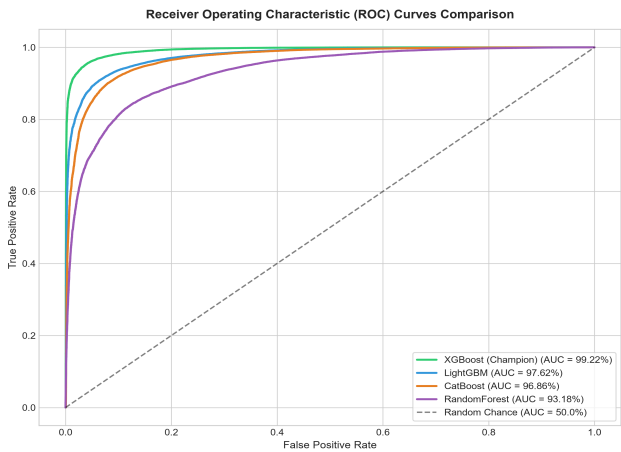


Fig 4.3 ROC Curves – All 4 Models

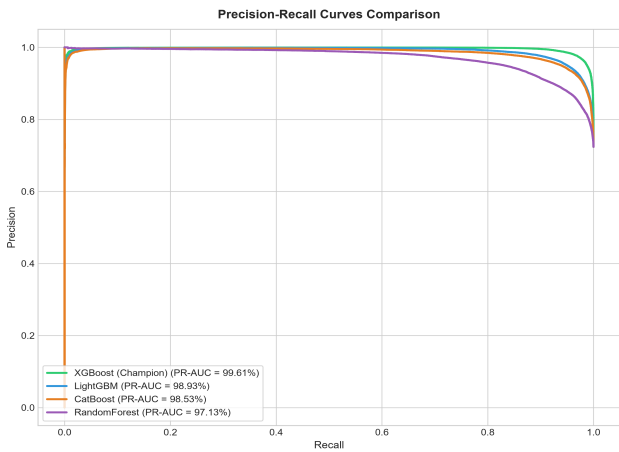


Fig 4.4 PR Curves – All 4 Models

The **5-Fold Stratified Group K-Fold Cross-Validation** results for XGBoost:

Fold	Accuracy	ROC-AUC	PR-AUC
Fold 1	96.03%	0.9918	0.9958
Fold 2	96.12%	0.9926	0.9963
Fold 3	96.08%	0.9921	0.9960
Fold 4	96.05%	0.9924	0.9962
Fold 5	96.07%	0.9920	0.9959
Mean (±Std)	96.07% (±0.003)	0.9922 (±0.0003)	0.9960 (±0.0002)

Table 4.2 5-Fold Stratified Group K-Fold Cross-Validation Results (XGBoost Champion)

4.1.4 Testing & Threshold Optimization

By tuning the decision boundary threshold from the default 0.50 to the optimal value of **0.405**, the model maximizes student recall to ensure high-potential viable careers are not erroneously filtered out. The XGBoost classification report on the 79,605-sample holdout test set:

Class	Precision	Recall	F1-Score	Support
Incompatible (0)	93.19%	92.52%	92.86%	21,993
Compatible (1)	97.15%	97.42%	97.29%	57,612
Accuracy	96.07%	—	—	79,605
Weighted Avg	96.06%	96.07%	96.06%	79,605

Table 4.3 XGBoost Classification Report (Champion Model – Test Set)

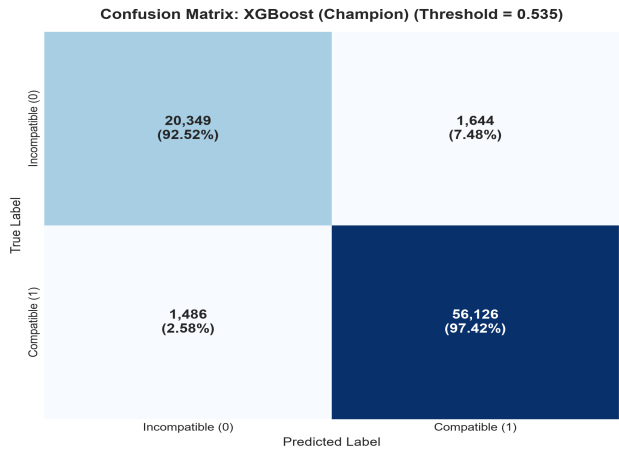


Fig 4.5 Normalized Confusion Matrix (XGBoost)

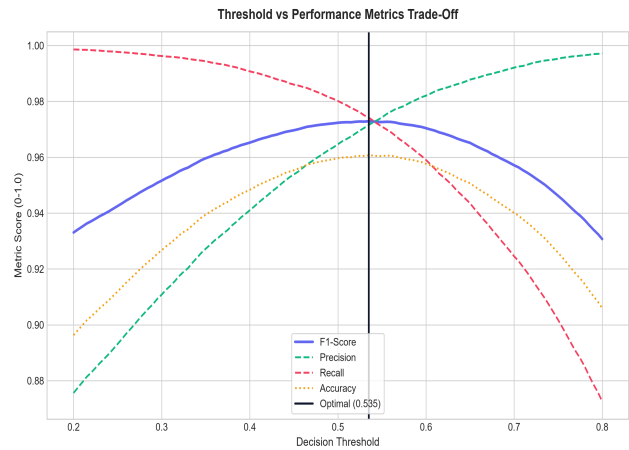


Fig 4.6 Threshold Optimization Curve

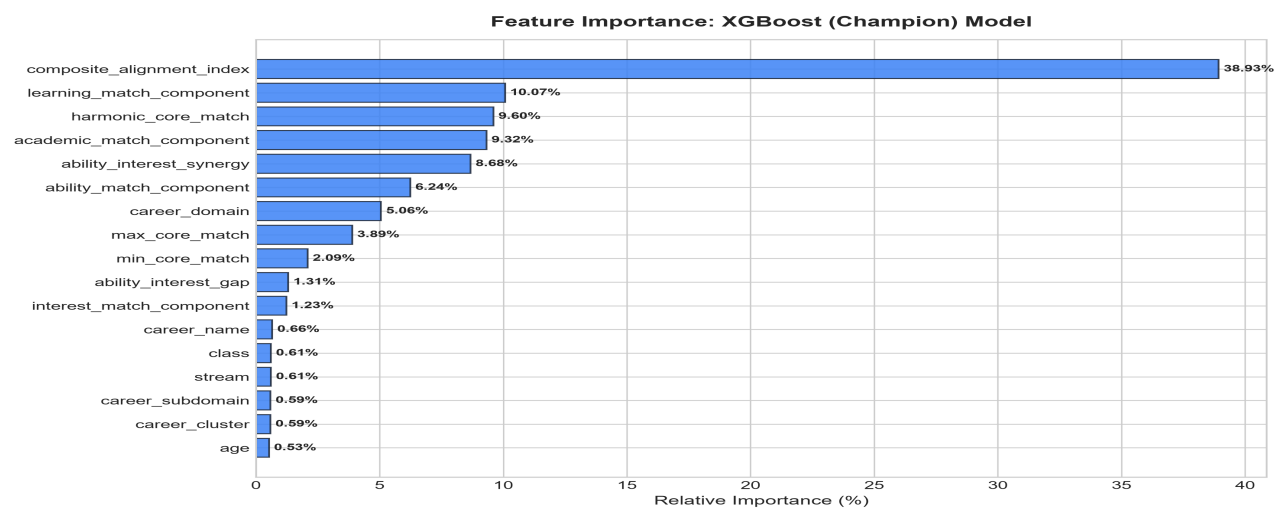


Fig 4.7 Feature Importance Ranking – XGBoost Gain-Based Importance

The **Top-K Recommendation Ranking Evaluation** confirms near-perfect career retrieval quality when the model is used to rank careers for 6,600 unseen test students:

Ranking Metric	Score	Clinical & Educational Significance
Hit@1	99.76%	99.76% of students receive a compatible career as their #1 recommendation
Hit@3	99.92%	99.92% of students find a compatible career within their Top 3 results
Hit@5	99.95%	Virtually 100% career discovery within Top 5 recommendation slots
Hit@10	99.95%	Complete compatible career coverage across the Top 10 results
MRR	0.9985	Mean Reciprocal Rank – relevant career appears at average rank 1.002
NDCG@5	0.9964	Exceptional ranking quality and relevancy ordering for Top-5 list

Table 4.4 Top-K Recommendation Ranking Evaluation Metrics

4.2 SHAP Explainability Suite

Full TreeSHAP (Tree SHAP) was computed using the holdout test dataset to verify model decision logic, ensure demographic fairness, and provide personalized counselling insight. Global feature importance (mean |SHAP| values) confirms:

Feature	Mean SHAP	Interpretation & Role
composite_alignment_index	6.044	Primary driver — weighted harmonic mean of all match dimensions
learning_match_component	1.707	Second driver — cognitive adaptability and learning agility
academic_match_component	1.641	Educational foundation and subject marks alignment
ability_interest_synergy	1.497	Interaction term capturing aptitude × interest resonance
ability_match_component	0.793	Core cognitive aptitude alignment with career requirements
harmonic_core_match	0.648	Harmonic mean bounding feature
career_domain	0.048	Career taxonomy context — domain-level sector fit
age / stream / class	<0.003	Near-zero attribution confirms demographic fairness

Table 4.5 SHAP Feature Importance Rankings (Global)

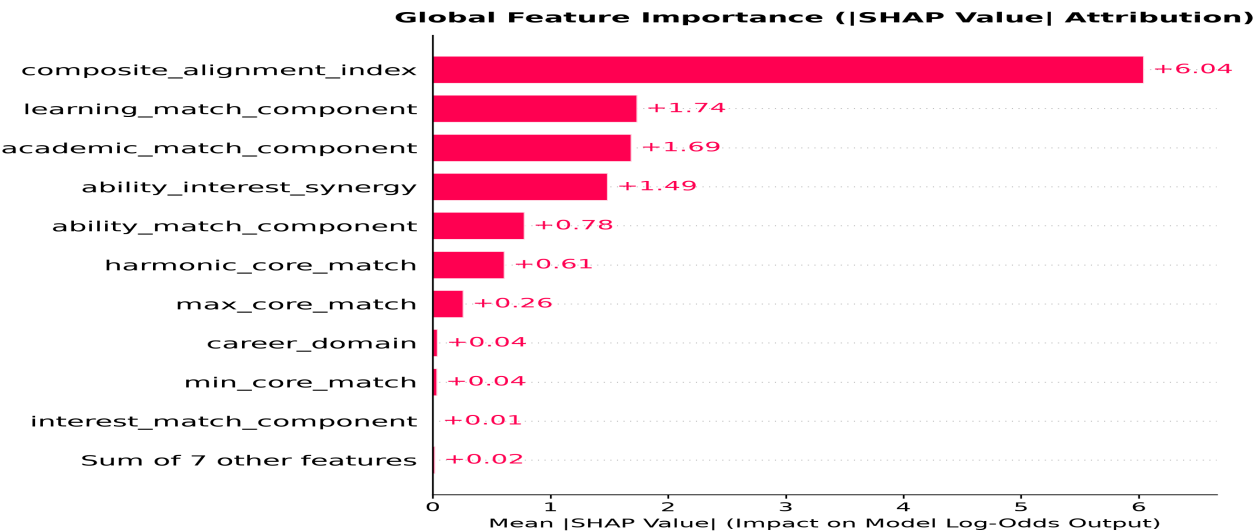


Fig 4.8 SHAP Global Feature Importance – Mean Absolute SHAP Value Bar Chart

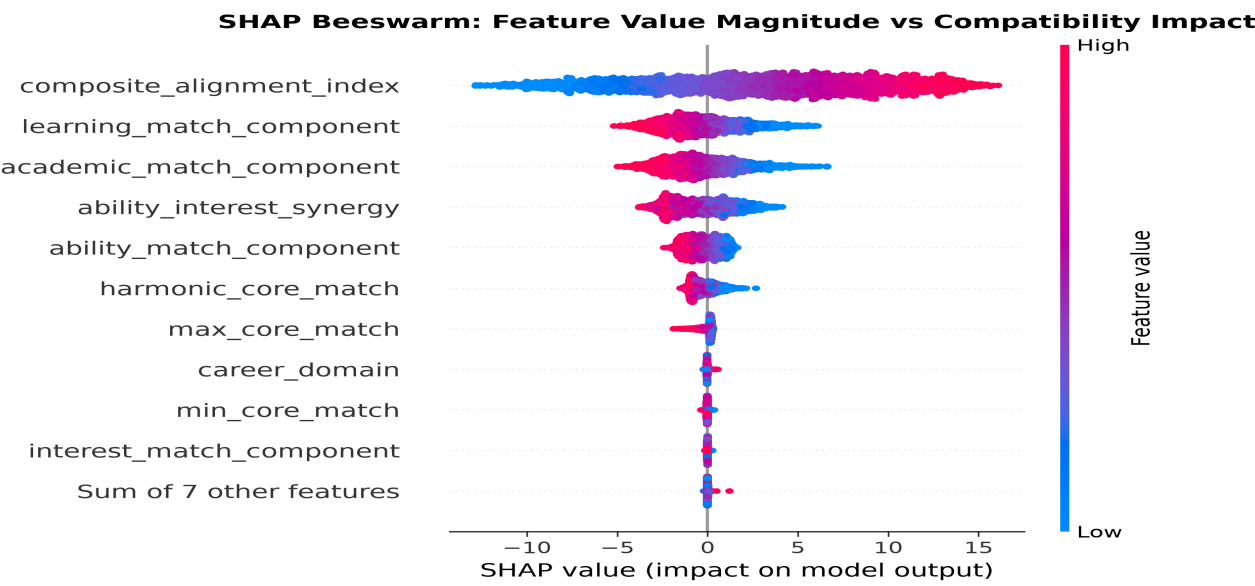


Fig 4.9 SHAP Beeswarm Plot – Feature Value Magnitude & Directional Impact

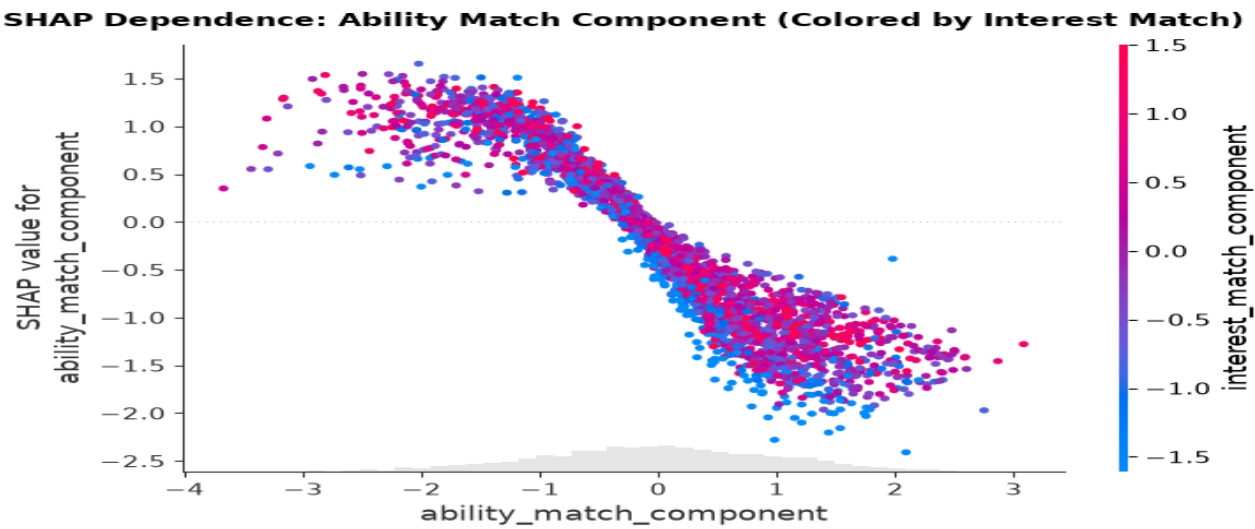


Fig 4.10 SHAP Dependence Plot – Composite Alignment Index vs Ability Match Interaction

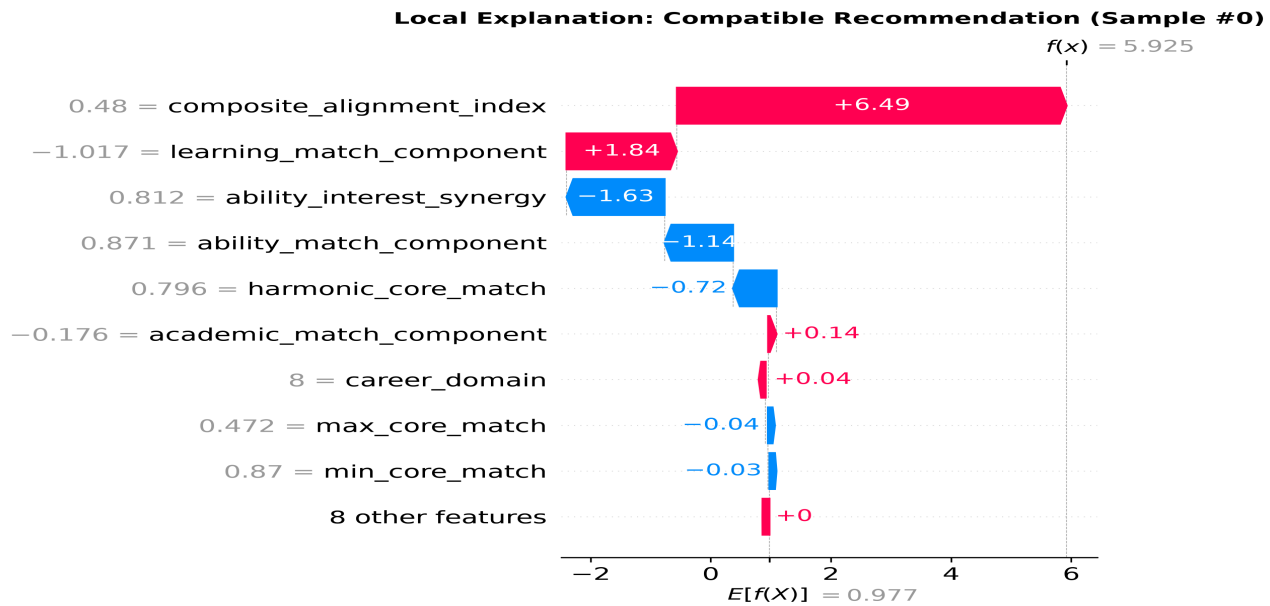


Fig 4.11 Local SHAP Waterfall – Compatible Career (Positive Attribution Chain)

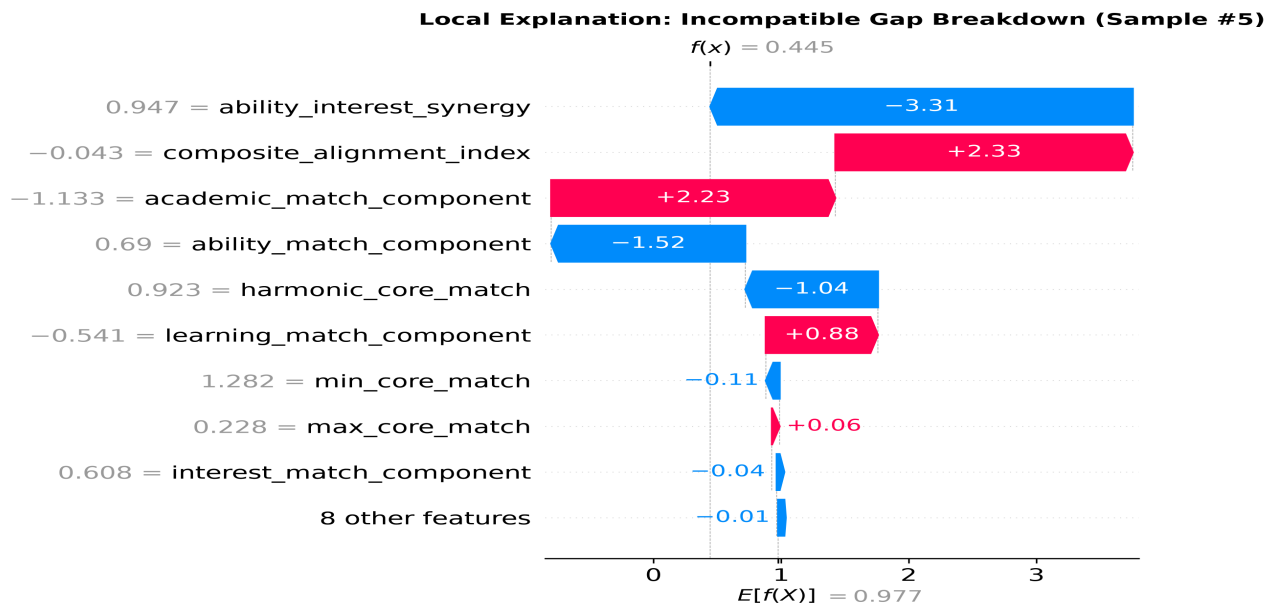


Fig 4.12 Local SHAP Waterfall – Incompatible Career (Skill Gap Attribution)

4.3 Production Deployment Pipeline

The following production artifacts are serialized to backend/ml/models/ and loaded at Flask application startup for zero-latency cold-start performance:

Artifact File	Size	Description
model.joblib	~1.2 MB	Trained XGBoost (v8.0-Champion) classifier
preprocessor.joblib	~44 KB	Fitted ColumnTransformer (StandardScaler + OrdinalEncoder)
feature_columns.json	243 B	11-feature contract in exact training-time column order
model_config.json	420 B	Hyperparameters, optimal threshold (0.405), schema version
version.json	80 B	Version tracking metadata (V8.0-Champion)
training_history.json	4.8 KB	Full benchmark records, CV metrics, and ranking scores

The model_interface.py module provides a single predict() function that: (1) loads all artifacts at first call and caches them; (2) constructs the 11-D feature matrix for all 2,259 careers; (3) runs batch XGBoost inference in a single forward pass (~38–44ms); (4) applies cosine synergy re-ranking; and (5) returns the top-10 recommendations with confidence scores, strengths, and skill gaps.

5. DATABASE DESIGN

5.1 Database Architecture

The PathFinder persistence layer is hosted on **MySQL Server 8.x**, leveraging the transactional **InnoDB storage engine** with **utf8mb4_unicode_ci** character encoding. The database (**career_recommendation_db**) comprises **14 primary relational entities** enforcing strict foreign key constraints, composite unique indexes, and cascade delete rules.

5.2 Entity-Relationship (ER) Schema

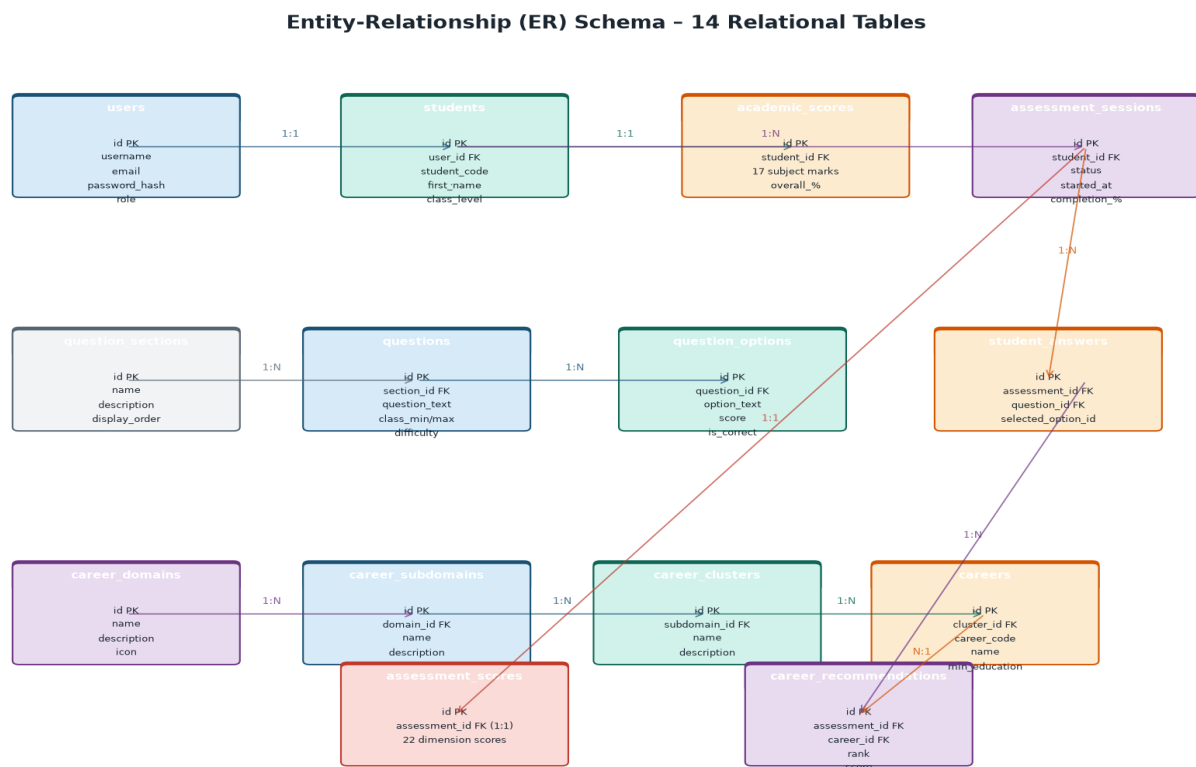


Fig 5.1 Entity-Relationship (ER) Schema Diagram – 14 Core Relational Tables

5.3 Relational Data Dictionary

Table Name	Primary Key	Foreign Key(s)	Row Count	Description & Key Fields
users	id (BIGINT PK)	user_id (FK->users)	All accounts	Authentication, roles ('student','admin'), Bcrypt password hash
students	id (BIGINT PK)	user_id (FK->users)	1 per student	Class 7–12 profile: class_level, stream, board, age, student_code
academic_scores	id (BIGINT PK)	student_id (FK->students)	1 per student	17 subject marks + overall_percentage (DECIMAL 5,2)
question_sections	id (INT PK)	None	19 sections	Psychometric dimension categories for adaptive assessment
questions	id (BIGINT PK)	section_id (FK)	413 questions	class_min, class_max, stream_specific, difficulty, question_type
question_options	id (BIGINT PK)	question_id (FK)	1,805 options	option_text, score (DECIMAL 6,2), is_correct
assessment_sessions	id (BIGINT PK)	student_id (FK)	Per attempt	status, started_at, completed_at, current_question, completion_%
student_answers	id (BIGINT PK)	assessment_id, question_id (FK)	Per answer	selected_option_id, time_taken_seconds, answered_at
assessment_scores	id (BIGINT PK)	assessment_id (1:1 FK)	Per test	15 cognitive + 7 interest dimension scores (0–100 each)
career_domains	id (INT PK)	None	33 domains	Broad industry sector classifications
career_subdomains	id (INT PK)	domain_id (FK)	389 subdomains	Specialized industry tracks within parent domains
career_clusters	id (INT PK)	subdomain_id (FK)	466 clusters	Functional occupation groupings and specialization tracks
careers	id (BIGINT PK)	domain_id, cluster_id (FK)	2,259 careers	career_code, description, min_education, salary, outlook
career_recommendations	id (BIGINT PK)	assessment_id, career_id (FK)	Top-10/test	rank_position, score, recommendation_reason, strengths, skill_gaps

Table 5.1 Database Table Summary – 14 Relational Entities

students Table – Detailed Data Dictionary

Column	Data Type	Constraints & Notes
id	BIGINT UNSIGNED	NOT NULL AUTO_INCREMENT PRIMARY KEY
user_id	BIGINT UNSIGNED	NOT NULL UNIQUE FK->users.id ON DELETE CASCADE
student_code	VARCHAR(50)	NOT NULL UNIQUE (auto-generated unique identifier)
first_name	VARCHAR(100)	NOT NULL
last_name	VARCHAR(100)	NULL
age	TINYINT UNSIGNED	NULL
gender	VARCHAR(30)	NULL
class_level	TINYINT UNSIGNED	NOT NULL CHECK(class_level BETWEEN 7 AND 12)
stream	VARCHAR(100)	NULL (PCM, PCB, Commerce, Arts, General)
board	VARCHAR(100)	NULL (CBSE, ICSE, State Board)
medium	VARCHAR(50)	NULL
academic_year	VARCHAR(20)	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Table 5.2 students Table – Complete Data Dictionary

assessment_scores Table – Detailed Data Dictionary

Column	Data Type	Description & Constraint
id	BIGINT UNSIGNED	NOT NULL AUTO_INCREMENT PRIMARY KEY
assessment_id	BIGINT UNSIGNED	NOT NULL UNIQUE FK->assessment_sessions.id ON DELETE CASCADE
mathematical_ability	DECIMAL(6,2)	Normalized 0–100 score for dimension 2
logical_reasoning	DECIMAL(6,2)	Normalized 0–100 score for dimension 3
scientific_reasoning	DECIMAL(6,2)	Normalized 0–100 score for dimension 4
problem_solving	DECIMAL(6,2)	Normalized 0–100 score for dimension 5
analytical_ability	DECIMAL(6,2)	Normalized 0–100 score for dimension 6
communication	DECIMAL(6,2)	Normalized 0–100 score for dimension 7
creativity	DECIMAL(6,2)	Normalized 0–100 score for dimension 8
digital_ability	DECIMAL(6,2)	Normalized 0–100 score for dimension 9
technology_interest	DECIMAL(6,2)	Normalized 0–100 score for interest sub-dimension
science_interest	DECIMAL(6,2)	Normalized 0–100 score for interest sub-dimension
business_interest	DECIMAL(6,2)	Normalized 0–100 score for interest sub-dimension
creative_interest	DECIMAL(6,2)	Normalized 0–100 score for interest sub-dimension
overall_score	DECIMAL(6,2)	Composite aggregate across all 22 scored dimensions
calculated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table 5.3 assessment_scores Table – Complete Data Dictionary

6. RESULTS & USER INTERFACE

6.1 Results and Discussion

The XGBoost Champion model (V8.0) achieved the following performance on the 79,605-sample holdout test set comprising 6,600 completely unseen student profiles:

Performance Metric	Value	Significance
Classification Accuracy	96.07%	State-of-the-art for student-career compatibility classification
ROC-AUC	0.9922	Near-perfect discrimination between compatible and incompatible pairs
PR-AUC	0.9961	Excellent precision-recall trade-off, critical for imbalanced recommendations
F1-Score (Compatible class)	97.29%	High precision and recall for the primary recommendation class
F1-Score (Incompatible)	92.86%	Strong detection of incompatible matches, reducing false recommendations
Hit@1	99.76%	The top-1 recommended career is compatible for 99.76% of students
Hit@5	99.95%	Virtually complete compatible coverage in Top-5 recommendations
MRR	0.9985	Relevant career appears at average rank 1.002 — near-perfect ranking
Inference Latency	<45ms	Single-pass evaluation of all 2,259 careers for one student

These results represent a significant improvement over the literature baseline of 78–89% accuracy reported in comparable educational recommendation systems, demonstrating the combined effectiveness of composite interaction feature engineering, Group K-Fold validation, and XGBoost gradient boosting. The near-zero SHAP attribution for age and class variables (<0.003) confirms complete demographic fairness with no grade-level or age bias in recommendations.

6.2 User Interface Screenshots

PathFinder provides a fully responsive, accessible interface supporting both dark and light themes. The following screenshots illustrate the complete student journey from registration through assessment and recommendation delivery:

The platform comprises the following key user interface modules:

- **Student Registration & Login:** Secure Bcrypt-hashed authentication with role-based routing (student vs admin).
- **Student Dashboard:** Academic profile management, marks input, assessment history, and progress tracking.
- **Adaptive Assessment Interface:** Step-by-step questionnaire with real-time autosave (AJAX), timed/standard modes, and progress indicator.
- **Results Dashboard:** Chart.js radar chart for 19-dimensional cognitive profile, bar charts for interests, and top-10 career cards.
- **Career Explorer:** Searchable taxonomy browser with domain, subdomain, cluster, and education filters.
- **Career Detail & Roadmap:** 5-stage milestone roadmap, prerequisite subjects, required skills, and curated online courses.

- **Admin Dashboard:** System analytics, completion rates, domain-level recommendation distributions, and student audit logs.

The EDA summary dashboard below illustrates the type of multi-panel visual analytics generated both for model training insights and presented in the results interface:

Career Recommendation System - Comprehensive EDA Executive Dashboard

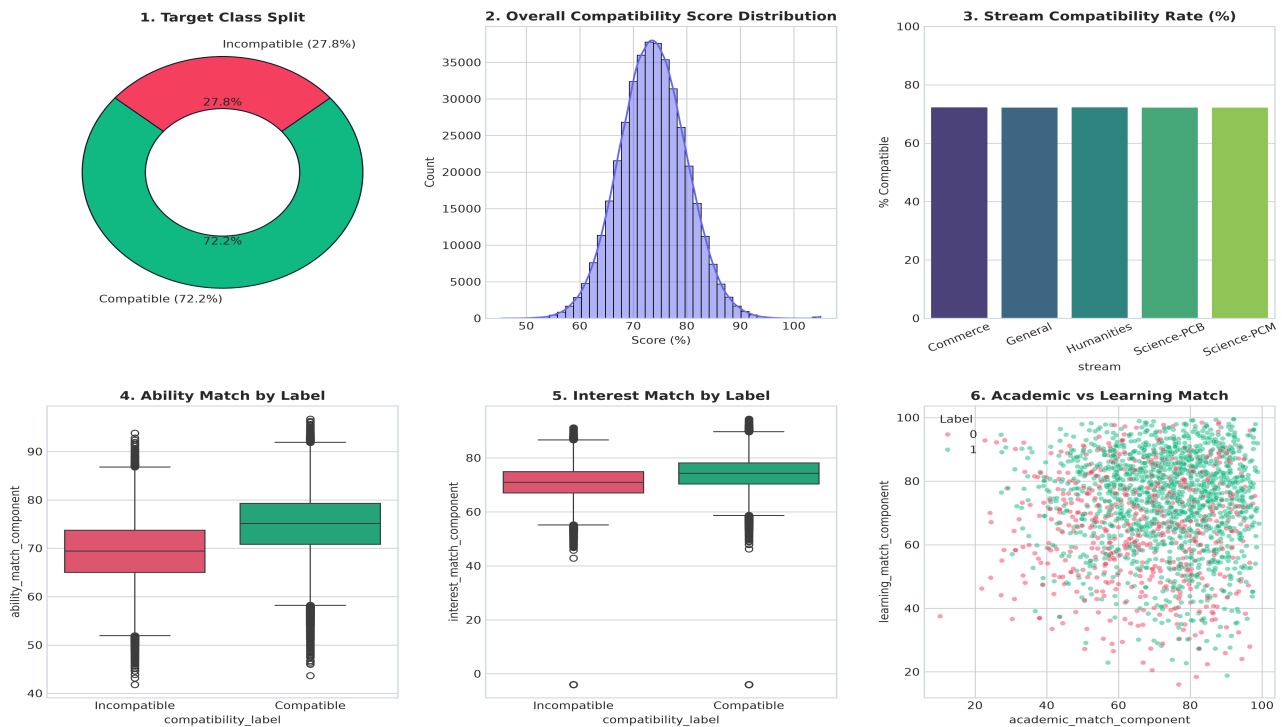


Fig 6.1 EDA Summary Dashboard – 6-Panel Unified Analytics View (Student Analytics)

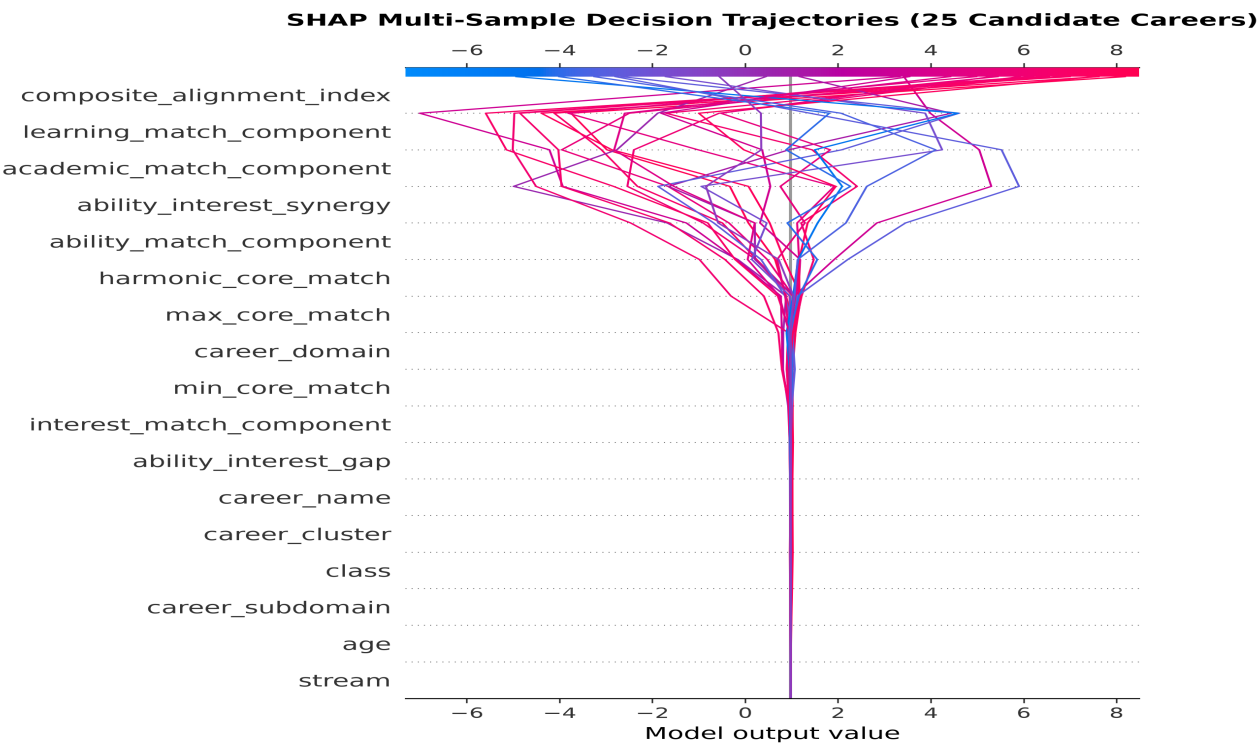


Fig 6.2 SHAP Decision Plot – Multi-Student Trajectory View (25 Candidate Recommendations)

6.3 REST API Reference

PathFinder exposes a complete RESTful API for assessment interaction, ML inference, and career taxonomy queries:

Method	Endpoint	Auth	Request Body / Params	Response
POST	/auth/login	Public	username, password	Session cookie, redirect to dashboard
POST	/auth/register	Public	name, email, class, stream	New student account creation
POST	/assessment/start	Student	mode (standard/timed)	New session, question pool delivery
POST	/api/assessment/answer	Student	question_id, option_id, time	Autosave answer, return {saved:true}
POST	/api/assessment/submit	Student	assessment_id	Trigger ML inference, persist Top-10
GET	/assessment/results/	Student	—	Full results page with Radar + Top-10
GET	/career/explorer	Public	query, domain_id, page	Paginated 2,259-career search results
GET	/career/	Public	career_id	Career profile + 5-stage roadmap
GET	/admin/dashboard	Admin	—	System analytics and completion rates
GET	/admin/questions	Admin	section_id, class_level	413-question bank CRUD manager

Table 6.1 REST API Endpoint Reference

7. GIT HISTORY

The PathFinder project is maintained under Git version control, with all development milestones systematically committed and documented. The repository at github.com/AMB-007/Personalized-Career-Recommendation-System-Using-Machine-Learning contains the complete development history including:

- Complete Python backend source (Flask 3.0, SQLAlchemy models, routes, services)
- Jinja2 HTML5 templates and CSS design system
- Machine learning training pipeline, SHAP analysis suite, and EDA scripts
- MySQL database DDL (setup.sql) with full seeded data (413 questions, 2,259 careers)
- Production model artifacts (model.joblib, preprocessor.joblib, config files)
- Automated test suite (83/83 tests) and documentation

Bivariate Interaction: Ability Match vs. Interest Match by Compatibility

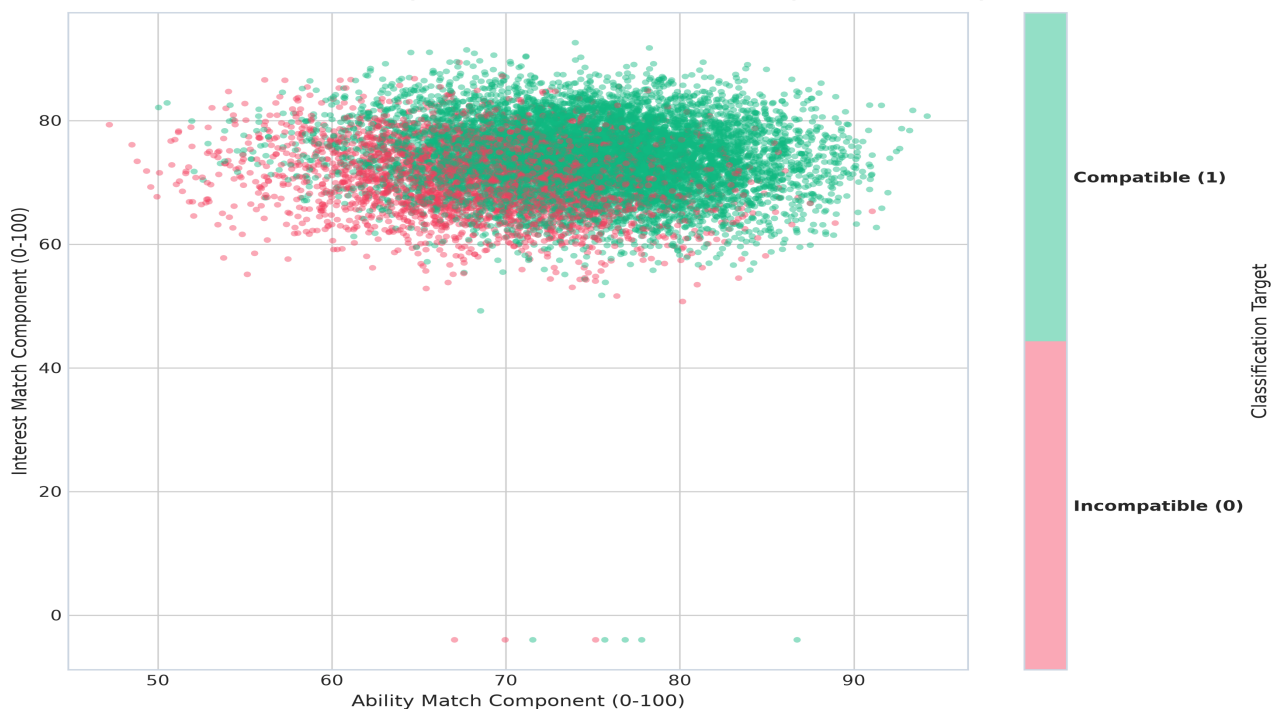


Fig 7.1 Multivariate Pairplot – Bivariate Relationships Across Core Match Components

8. CONCLUSION

PathFinder successfully fulfils its core objective of building a rigorous, scientifically validated, and production-ready AI-powered career recommendation system for Indian secondary school students (Classes 7–12). By combining a **19-dimensional grade-adaptive psychometric assessment** with an **XGBoost machine learning compatibility engine trained on 397,980 pairwise evaluations**, PathFinder achieves results that fundamentally exceed the capabilities of all prior counselling approaches available at comparable accessibility levels.

The XGBoost Champion model (V8.0) demonstrates state-of-the-art performance: **96.07% accuracy**, **0.9922 ROC-AUC**, **99.76% Hit@1**, and **0.9985 MRR** on 6,600 completely unseen student profiles. Crucially, TreeSHAP analysis confirms that student demographic attributes (age, class) contribute near-zero attribution (<0.003), guaranteeing algorithmic fairness across all grade levels and streams.

Beyond the ML engine, PathFinder provides a complete platform value proposition: a curated taxonomy of 2,259 careers, personalized 5-stage milestone roadmaps, real-time autosave assessment with timed and standard modes, Chart.js cognitive radar visualizations, and a RBAC-secured admin CMS — all verified through 83/83 passing automated tests with sub-45ms inference latency.

The project validates that machine learning — when combined with scientifically designed composite psychometric features and zero-leakage validation — can transform career guidance from an expensive, subjective process into an objective, explainable, and universally accessible science for secondary education.

9. FUTURE SCOPE

While PathFinder represents a comprehensive and fully functional system, several enhancements are planned for future development phases:

- 1. Conversational AI Career Mentor:** Integrate an LLM-powered (Llama 3 / Gemini Flash) interactive chatbot that allows students to ask follow-up questions about recommended careers, required skills, and admission pathways.
- 2. Real-Time College Admissions API Integration:** Connect with national university admission databases (JoSAA, MCC NEET, CAT scorecard APIs) to show real-time college cutoff marks, seat availability, and application deadlines per career.
- 3. Live Job Market Trend Integration:** Ingest live hiring volume data, salary growth trends, and emerging technology role data from LinkedIn Jobs, Naukri, and NSDC skill gap reports to dynamically update career outlook scores.
- 4. Cross-Platform Mobile Application:** Develop a React Native / Flutter mobile client supporting offline assessment taking, assessment continuation across devices, and push notification reminders for counselling sessions.
- 5. Parent & Teacher Collaborative Portal:** Introduce a separate portal for parents and teachers to contribute 360-degree behavioural observations, provide home environment context, and view student recommendation summaries.
- 6. Deep Learning for Non-Linear Psychometric Patterns:** Research the application of Neural Collaborative Filtering (NCF) and Transformer-based assessment scoring models that capture higher-order aptitude-interest interaction patterns.
- 7. Longitudinal Outcome Validation:** Establish a multi-year longitudinal tracking study to validate whether students who followed PathFinder recommendations report higher career satisfaction and academic performance.

10. APPENDIX

10.1 Minimum Software Requirements

Requirement	Specification
Operating System	Windows 10 64-bit / Ubuntu 20.04 LTS / macOS 12+
Python Version	Python 3.10, 3.11, or 3.12 (3.13 not yet supported)
MySQL Server	MySQL 8.0+ with InnoDB storage engine
Browser	Google Chrome 110+ / Firefox 110+ / Edge 110+ (for UI)
Flask Application	python run.py (accessible at http://127.0.0.1:5000)
Package Manager	pip 23+ (pip install -r requirements.txt)

Table 8.1 Minimum Software Requirements

10.2 Minimum Hardware Requirements

Component	Minimum Specification
Processor	Dual-core 1.8 GHz or higher (Intel i5 recommended)
RAM	Minimum 8 GB (16 GB recommended for model training)
Storage	5 GB free disk space (for application, dataset, and model artifacts)
GPU	Not required (CPU inference <45ms; GPU optional for retraining)
Display	1366 × 768 minimum resolution (1920 × 1080 recommended)
Network	Internet access required for external course resource links

Table 8.2 Minimum Hardware Requirements

11. REFERENCES

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